Decoding Risk Factors of Alzheimer's Disease; Comparative Analysis of Classification Systems

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Introduction

This project focuses on decoding the risk factors of Alzheimer's disease to classify patients who have Alzheimer's from healthy patients based on demographic, lifestyle, and clinical variables. Alzheimer's is one of the most common neurodegenerative disorders, which has become a growing public health concern due to its progressive and irreversible nature. It primarily affects the patients through symptoms like memory loss, cognitive decline, and reduced ability to perform daily function. Currently, diagnosis is generally based on clinical assessments and cognitive testing, but there is growing interest in identifying early indicators that potentially signal an individual's risk of developing the disease in the future. Doing so can help focus clinical screenings by understanding the interaction of various risk factors, which can significantly help alleviate symptoms and slow the progression of Alzheimer's in patients. In this study, we examine whether Alzheimer's can be classified using a combination of demographic, lifestyle, and clinical variables using a machine learning classifier model. Specifically, we focus on understanding whether risk factors can meaningfully distinguish between individuals with Alzheimer's and those without it and how different classification models perform.

The central problem we are trying to investigate is whether non-diagnostic variables, such as lifestyle factors and medical history, can be reasonably and accurately used to classify Alzheimer's. Using a dataset of over 2,000 patients containing both individuals diagnosed with Alzheimer's and individuals who are healthy as controls, we applied supervised learning methods to determine how well these variables can predict the diagnosis. A successful model that can predict risk rather than just diagnose a patient based on symptoms can be particularly important. This is due to the current nature of Alzheimer's symptoms, diagnosis, and its

dependence on cognitive assessments such as memory tests and functional evaluations. These typically only indicate Alzheimer's well into its onset, making it difficult to minimize or slow significant damage to the brain and neurons. While symptom measures can produce high classification accuracy, they cannot necessarily help identify individuals at risk before these symptoms are noticeable. Consequently, this study separates the analysis into two conditions: one that includes all available variables in the dataset and one that excludes all symptom-related variables and focuses on risk factors.

However, this approach poses challenges for several reasons. First, Alzheimer's is influenced by factors that have a complex, non-linear relationship and only a weak individual relationship with the disease. Most demographic and lifestyle variables such as age, BMI, or blood pressure have little direct correlation with Alzheimer's diagnosis when looking at them individually. This makes it very difficult for machine learning models to identify a clear pattern when relying only on these variables and not a combination of them. Second, there is a significant overlap between individuals with Alzheimer's and individuals without Alzheimer's across these risk factors, further complicating classification. And lastly, since some variables in the dataset represent symptoms rather than risk predictors, it is important to accommodate the dataset accordingly. Including all variables will likely inflate model performance and stray away from predicting risk toward reproducing an already established diagnosis. Thus, it is necessary to differentiate between risk and symptom variables to ensure that the classification model focuses on the problem of early risk detection.

Understanding this problem matters for both clinical and analytical reasons. From a healthcare perspective, earlier identification of individuals at risk for Alzheimer's disease could lead to better intervention strategies, improved medication, and more efficient allocation of

medical resources. If meaningful patterns could be detected using risk factors, such as lifestyle or medical history, clinical screenings could be more focused. However, if these risk factors cannot sufficiently be classified, it indicates that current knowledge of risk factors is not enough for predictive models and more research with better data is necessary. From an analytical perspective, we are able to evaluate how different machine learning methods perform under realistic conditions. In many real-world datasets, the strongest predictors are not always available or may not be appropriate for the intended goal. This study allows us to compare model performance when using variables such as cognitive assessments and lifestyle factors. It also allows us to examine how techniques like dimensionality reduction impact classification accuracy. These comparisons are important for understanding not only which models perform best, but also why they perform the way they do.

Another goal of this study is to compare the effectiveness of different machine learning approaches in classifying Alzheimer's disease under varying conditions. We evaluate two main classification methods: decision trees and support vector machines (SVM). Also, we investigate whether applying locally linear embedding (LLE), a dimensionality reduction technique, improves model performance. By testing combinations of these methods, we aim to determine how model choice and data preprocessing influence classification outcomes. Additionally, this study aims to assess whether Alzheimer's disease can be reliably classified using risk factors alone, without relying on direct symptom indicators. Next, it aims to compare the performance of different algorithms, including decision trees and SVM. Lastly, it is evaluating if reducing the dimensionality of the dataset will improve or hinder classification performance.

Based on our understanding of the dataset and the nature of Alzheimer's disease, we anticipated that models trained on the full dataset, including both risk factors and symptom

measures, would achieve higher classification accuracy due to the strong relationship between the variables and diagnosis. On the other hand, we anticipated that models trained only on risk factors would perform worse, given the weaker associations observed earlier. We also hypothesized that dimensionality reduction might improve performance by simplifying the dataset and reducing noise, particularly given the relatively large number of variables.

Overall, this study aims to examine not only how well Alzheimer's disease can be classified, but also which types of variables contribute meaningfully to that classification. By comparing different machine learning modeling approaches, we are evaluating the potential and the limitation of machine learning in Alzheimer's disease prediction.

Data and Preprocessing

Where the Data Comes From

For this project, we used the Alzheimer's Disease Patient Dataset published by Globose Technology Solutions, which we accessed through Kaggle. The dataset contains 2,149 patient records and 32 columns covering demographic information, lifestyle factors, medical history, clinical measurements, cognitive assessments, symptoms, and a binary Alzheimer's diagnosis label. Of the 2,149 patients, roughly 35.4% (759) have a positive Alzheimer's diagnosis and 64.6% (1,390) are healthy controls, giving us a moderately imbalanced classification problem. We selected this dataset because it pairs risk factors (e.g., lifestyle, cardiovascular disease, family history) with cognitive and functional symptoms in the same record, allowing us to model both early risk factors and clinical presentation within a single supervised learning framework.

Key Variables

The dataset included 32 variables organized into six predictor domains and one outcome variable. Predictor variables covered demographics, medical history, clinical measurements , cognitive and functional assessments, and symptoms. The outcome variable was diagnosis, coded as 0 = healthy and 1 = Alzheimer's disease. The full list of variables included in the dataset is presented in Table 1.

Cleaning and Preprocessing

Fortunately, the dataset is already very clean. We confirmed that there are no missing values in any column, so no imputation was required. We dropped two columns that carried no predictive information: PatientID (a row identifier) and DoctorInCharge (a constant placeholder string). We used one-hot encoding on the Ethnicity variable to make it suitable for scale sensitive analysis. All other categorical variables are already dummy-coded as integers. We inspected the continuous variables for outliers using the IQR rule (values below Q1 − 1.5*IQR or above Q3 + 1.5*IQR). No outliers were detected. For the LLE-based pipelines, we applied StandardScaler (mean = 0, SD = 1) to the continuous and ordinal variables only, leaving binary indicators on their original 0/1 scale. For the plain decision-tree pipeline, no scaling was applied because tree splits are scale-invariant.

Finally, because several cognitive variables function as symptoms rather than independent risk factors, we ran our models on two feature sets: (1) the full dataset, (2) the full dataset excluding symptom columns (MMSE, FunctionalAssessment, MemoryComplaints, BehavioralProblems, ADL, Confusion, Disorientation, PersonalityChanges, DifficultyCompletingTasks, and Forgetfulness). All splits used a 70/30 train/test split, stratified by Diagnosis with random_state = 42 for reproducibility.

Exploratory Data Analysis

Participants are, on average, 75 years old with a standard deviation of 9 years. The sample is evenly distributed by gender, with 50% female and 50% male participants. Most participants are Caucasian (60%), followed by African American (20%), Asian (10%), and Other ethnicity (10%). Education levels vary, with the largest group completing high school (39.7%), followed by bachelor's degree (29.6%), no formal education (20.8%), and higher education (9.9%) (see Table 2).

Continuous variables

The dataset contains 15 continuous variables covering lifestyle (BMI, alcohol consumption, physical activity, diet quality, sleep quality), clinical measurements (blood pressure, cholesterol panels), and cognitive/functional assessments (MMSE, Functional Assessment, ADL). EducationLevel, though ordinal, is also treated as continuous for scaling purposes.

Descriptive Statistics

As shown in Table 3, we have 15 continuous variables, mean, standard deviation, and range are listed in the table. Education level is later used as continuous variables for it is an ordinal variable that can be used for scaling.

Distribution of Continuous Variables

Histograms across all 15 continuous variables reveal broadly uniform or near-uniform distributions, as indicated in Figure 1.

KDE Overlay by Diagnosis Group

To assess which continuous variables carry predictive signals, kernel density estimates (KDE) were plotted separately for Diagnosis = 0 (healthy) and Diagnosis = 1 (Alzheimer's), visualized in Figure 2. Most lifestyle and clinical variables show nearly identical distributions

between groups, suggesting limited individual predictive power. Three variables stand out with clear diagnostic separation: MMSE scores are substantially lower in the Alzheimer's group (concentrated near 0–10) versus healthy patients (broadly distributed up to 30); Functional Assessment and ADL scores follow the same pattern — Alzheimer's patients cluster toward lower functional scores.

Categorical Variables

The dataset contains 17 categorical variables. Categorical variables include gender, binary medical history (smoking, diabetes, depression, family history of Alzheimer's, cardiovascular disease, hypertension, head injury) and symptom indicators (memory complaints, behavioral problems, confusion, disorientation, personality changes, difficulty completing tasks, forgetfulness), as well as ordinal/nominal variables Education Level and Ethnicity.

Binary Variables: Discrimination Gap

The plot below shows the difference in prevalence between Alzheimer's and healthy groups for each binary variable (P(value =1 | Alzheimer's) − P(value =1 | Healthy)). For each variable, a value of 1 indicates the presence of that attribute or condition. For example, for MemoryComplaints, a value of 1 means the individual reports having memory complaints. Positive values (shown in red) indicate that the condition is more prevalent among individuals with Alzheimer's than among healthy controls. Larger positive values therefore reflect stronger positive associations with Alzheimer's diagnosis. Negative values (shown in green) indicate that the condition is more prevalent among healthy individuals.

As shown in Figure 3, Memory Complaints and Behavioral Problems show the largest positive gaps: patients with Alzheimer's report these symptoms at substantially higher rates than healthy individuals. Most demographic and medical history (smoking, family history, diabetes,

cardiovascular disease) show differences near zero, indicating limited discriminatory value on their own.

Non-binary Variables: Alzheimer's Rate by Level

For Education Level and Ethnicity, the Alzheimer's diagnosis rate within each category level is shown relative to the overall population rate (35.4%). Positive values (shown in red) indicate that individuals in that category have a higher-than-average Alzheimer's diagnosis rate, while negative values (shown in green) indicate a lower-than-average diagnosis rate.

As demonstrated in Figure 4, Education Level shows a negative gradient: patients with higher education tend to have slightly lower Alzheimer's rates. Ethnicity shows Asian has the highest rate of Alzheimer's across all groups, after accounting for the overall Alzheimer's diagnosis rate.

Correlation with Diagnosis

Pearson correlations between all features and Diagnosis confirm the visual patterns above. The five strongest correlates are FunctionalAssessment (-0.365), ADL (-0.332), MemoryComplaints (0.307), MMSE (-0.237), and BehavioralProblems (0.224) (see Table 4).

All remaining variables, including age, BMI, blood pressure, cholesterol, and all demographic variables, correlate with diagnosis at $|r| < 0.06$, effectively near zero. The dataset's predictive signal is almost entirely concentrated in cognitive/functional assessment scores and symptoms. Demographic, lifestyle, medical history, and clinical biomarkers show negligible association with diagnosis.

Methods

Based on the exploratory data analysis, symptom indicators, such as cognitive/functional variables (MMSE, Functional Assessment, ADL), were strongly related to diagnosis. Therefore, we trained two models: one using only risk factors, and thereby excluding the ten highly-correlated diagnostic variables, and one using the full dataset. This allowed us to compare when Alzheimer's is classified from risk factors alone versus when diagnostic indicators are included. To understand model performance, we analyzed the confusion matrix, encompassing true positives, true negatives, false positives and false negatives. As an extension to these preliminary results, we compared the precision, recall, f-1 score and support metrics. Precision indexes the accuracy of positive predictions, while recall indexes the accuracy of detecting true cases. F-1 score represents the harmonic mean of the prior metrics of precision and recall, while support represents the number of actual instances in each class . Lastly, we investigated model accuracy, representing overall performance of the classifier, as well as ROC AUC. The ROC AUC curve plots true positives against true positives (y-axis) against false positives (x-axis) to index the model's ability to separate classes across different probability thresholds (Fawcett, 2006).

Linear Local Embedding (LLE)

Locally Linear Embedding (LLE) is an unsupervised dimensionality reduction algorithm. This algorithm follows three steps: first, identify the k-nearest neighbors of each data point; second, compute reconstruction weights using linear combinations of those neighbors; and third, map the data to a lower-dimensional space using those weights. This effectively maps higher-dimension data into lower-dimensional space by identifying non-linear structures in the dataset (Roweis & Saul, 2000). This algorithm was specifically leveraged due to its ability to

identify non-linear structures, as opposed to data-reduction algorithms such as Principal Components Analysis (PCA; Jolliffe, 2002) that only identify linear structures.

Moreover, since our analytic dataset consisted of 32 variables, we implemented LLE to determine whether prior data-reduction aids downstream classification algorithms. In our implementation, preprocessing steps differed relative to the LLE algorithm. Since LLE utilizes Euclidean distances to find the k nearest neighbours of a data point (van der Maaten et al., 2009), it is sensitive to variables' scales. Thus, one copy of the dataset underwent scaling and subsequent LLE data reduction, while another copy was retained in its original form (i.e. non-scaled). Fixed values of 5 dimensions and 15 neighbours were chosen to allow sufficient variables for the Decision Tree algorithm.

Decision-Tree

A Decision Tree (DT) is a supervised classification model that classifies cases by repeatedly splitting the data based on the feature and threshold that best separate the target groups. Each case moves from the root node to a final leaf node, where the predicted class is based on the majority class in that leaf.

We used a Decision Tree classifier with the default Gini impurity criterion. The data were split into 70% training and 30% test sets using stratified sampling to preserve class proportions. To reduce overfitting, we performed hyperparameter tuning using 5-fold cross-validation on the training set, optimizing for ROC-AUC. The grid search included tree depths of {2, 3, 5}, minimum samples per leaf of {5, 10, 20}, and class weights {None, "balanced"} to account for class imbalance. The best-performing model was then evaluated on the held-out test set.

Support Vector Machine

The Decision-Tree classification algorithm was comparatively evaluated against the Support Vector Machine (SVM) algorithm. SVM is a supervised classification algorithm that maximizes the distance between each class in higher-dimensional space by determining an optimal 'hyperplace,' otherwise understood as a boundary. This model identifies support vectors (i.e. data points that are closest to the boundary) to define the boundary (Cortes & Vapnik, 1995). Since we have 32 analytic variables, SVM was specifically leveraged due to its ability to handle complex, high-dimensional data. Moreover, SVM was selected because of its ability to identify both linear and non-linear boundaries (Cortes & Vapnik, 1995), since we expect non-linear relationships between variables in our dataset.

Preprocessing was also subjectively implemented relative to the SVM algorithm. As SVM utilizes distances to maximize a margin in higher-dimensionality, it is sensitive to variables' scales. Thus, all continuous and ordinal variables that were subjected to SVM were previously scaled. Then, SVM was implemented on one dataset that was scaled and not previously reduced by LLE, as well as another dataset that was both scaled and previously reduced by LLE. Moreover, within the SVM implementation, two different kernels were used. A linear kernel was tested to determine whether the sample could be classified by a straight line, whereas a Radial Basis Function (RBF) kernel was tested to determine whether a flexible, non-linear line could effectively classify our sample. Default hyperparameters were chosen (C = 1.0 to balance maximization of the margin and classification error, gamma = scale to adapt by feature variance for the RBF kernel. Class weights were applied for all SVM implementations, to account for the unequal proportion of people with and without Alzheimer's disease.

Results

Decision Tree

Risk Assessment: No LLE + Decision Tree. Using the risk-only dataset (10 symptom/cognitive variables excluded) without dimensionality reduction. The model achieved a ROC-AUC of 0.517 and overall accuracy of 52.1%, only marginally above chance (Table 5). The model classified healthy individuals with precision (0.65) but poor recall for Alzheimer's patients (0.46), meaning over half of true Alzheimer's cases were missed. The near-chance AUC (0.517) indicates the risk-factor variables alone carry insufficient discriminatory power. This confirms the exploratory analysis finding that lifestyle, demographic, and medical history variables have near-zero individual correlations with diagnosis, as visualized in Figure 5. The tree, shown in Figure 6, splits first on **Sleep Quality**, followed by health-related variables such as blood pressure and cholesterol. Despite capturing some structure, the splits do not lead to strong class separation.

Risk Assessment: LLE + Decision Tree. Applying LLE (5 components) prior to classification on the risk-only dataset did not improve performance. ROC-AUC is 0.515, with an overall model accuracy of 48.4%. This is demonstrated in the performance metrics and confusion matrix visualized in Table 6 and Figure 7.

Full Variables: No LLE + Decision Tree. Including all 32 variables, in particular the cognitive and functional assessment scores highly correlated with diagnosis, dramatically improved performance. The optimal tree depth was 5, achieving a ROC-AUC of 0.938 and accuracy of 93.8% (Table 7).

Both classes were classified with high precision and recall (≥0.91), and the weighted F1-score of 0.94 reflects consistently strong performance across the imbalanced classes. This

result is not surprising given that MMSE, Functional Assessment, and ADL are essentially proxies for Alzheimer's status. While impressive in prediction, this model contains limited practical value since these variables require clinical assessment and are themselves diagnostic indicators. The confusion matrix, feature importance and ROC AUC curve is visualized in Figure 8. The decision tree in this condition, and as shown in Figure 9, demonstrates that the most important predictors are:

- Functional Assessment (highest importance)
- MMSE (cognitive score)
- ADL (daily functioning)
- Behavioral Problems
- Memory Complaints

These variables dominate the model, while traditional risk factors (e.g., cholesterol, BMI, lifestyle variables) contribute minimally. The tree first splits on Functional Assessment, followed by MMSE and ADL, forming a clear hierarchy of cognitive and functional impairment: lower functional and cognitive scores related to more likely classified as Alzheimer's.

Full Variables: LLE + Decision Tree. Applying LLE to the full-variable dataset substantially degraded performance despite the rich input. The optimal depth was 3, with ROC-AUC of 0.581 and accuracy of 65.7%. The model became biased toward predicting Alzheimer's. The Decision Tree model achieved an overall accuracy of 0.66. However, performance differed substantially across classes. For healthy individuals, the model demonstrated high recall (0.97) and moderate precision (0.66), indicating that most healthy cases were correctly identified. In contrast, performance for Alzheimer's patients was poor, with a

recall of only 0.09, meaning that the majority of true Alzheimer's cases were misclassified as healthy. The confusion matrix and ROC AUC curve is visualized in Figure 10 for this condition.

In summary, the full-variable model without LLE achieves near-perfect classification but relies on clinically proximate indicators. Risk-only models, the more practically meaningful condition, perform near chance regardless of LLE. Demographic and lifestyle risk factors in this dataset carry insufficient signal for reliable Alzheimer's classification. (Table 8)

Support Vector Machine

Risk assessment: No LLE and SVM linear kernel. The Linear SVM was applied to the risk-focused dataset without prior LLE data reduction. As shown in Table 9, this model received an accuracy of 50.54%, indicating poor overall classification of our sample. Class-wide results showed that the model classified healthy individuals, marked as class 0 (F1-score = 0.56), better compared to Alzheimer's patients, marked as class 1 (F1-score = 0.43). This suggests a slight bias towards correct classification of the majority class of healthy individuals. The confusion matrix, shown in Figure 11, indicated that the model produces an abundance of misclassifications for both healthy individuals and Alzheimer's patients, visualized by the large number of false positives and false negatives. Moreover, the model misclassifies healthy individuals as patients with Alzheimer's more often than it correctly classifies them as healthy individuals. The relatively low recall for both healthy individuals and Alzheimer's patients, visualized as 0.49 and 0.54 respectively, suggests that the model is ineffective in distinguishing between individuals with or without diagnoses. Lastly, the ROC curve (Figure 12; ROC AUC = 52.27%) approximately follows the trend of the diagonal line, further confirming our prior findings of poor overall model performance.

Risk assessment: No LLE and SVM RBF kernel. The non-linear RBF SVM was applied to the risk-focused dataset without prior LLE data reduction. As shown in Table 9, this model received an accuracy of 47.75%, indicating poor classification of our sample. Class-wide results showed that the model classified healthy individuals, marked as class 0 (F1-score = 0.56), better compared to Alzheimer's patients, marked as class 1 (F1-score = 0.36). This suggests a bias towards correct classification of the majority class of healthy individuals. As shown in Figure 13, the confusion matrix indicated that the model produces an abundance of misclassifications for both healthy individuals and Alzheimer's patients, visualized by the large number of false positives and false negatives. The model is specifically poor at detecting true cases of Alzheimer's patients, with a low recall of 0.41. Lastly, the ROC curve (Figure 14; ROC AUC = 48.11%) approximately follows the trend of the diagonal line, often falling below. This further confirms our prior findings of poor overall model performance.

Risk assessment: LLE and SVM linear kernel. The Linear SVM was applied to the risk-focused dataset with prior LLE data reduction. As shown in Table 10, this model received an accuracy of 64.65%, indicating improved but poor overall classification of our sample. Class-wide results showed that the model classified healthy individuals, marked as class 0 (F1-score = 0.79), substantially better compared to Alzheimer's patients, marked as class 1 (F1-score = 0.00). This indicates a complete failure to identify individuals with Alzheimer's disease. The confusion matrix (Figure 15) showed that the model doesn't classify any individuals in our sample as having Alzheimer's disease, visualized by zero true positives and zero false positives. This is further reflected in our recall and precision metrics equating to zero, which further confirms our model's inability to identify individuals with Alzheimer's disease. Lastly,

the ROC curve (Figure 16; ROC AUC = 48.69%) approximately follows the trend of the diagonal line, further confirming our prior findings of poor overall model performance.

Risk assessment: LLE and SVM RBF kernel. The non-linear RBF SVM was applied to the risk-focused dataset with prior LLE data reduction. As shown in Table 11, this model received an accuracy of 50.08%, indicating near-chance classification of our sample. Class-wide results showed that the model classified healthy individuals, marked as class 0 (F1-score = 0.60), better compared to Alzheimer's patients, marked as class 1 (F1-score = 0.34). This suggests a bias towards correct classification of the majority class of healthy individuals. The confusion matrix (Figure 17) showed that the model produces an abundance of misclassifications for both healthy individuals and Alzheimer's patients, visualized by the large number of false positives and false negatives. Moreover, the model misclassifies patients with Alzheimer's more often than it correctly classifies them. The low recall for Alzheimer's patients (0.37) relative to healthy individuals (0.57), suggests that the model performs poorer in detecting individuals with Alzheimer's disease. Lastly, the ROC curve (Figure 18; ROC AUC = 48.41%) approximately follows the trend of the diagonal line, often falling below. This further confirms our prior findings of poor overall model performance.

Post-hoc Broad classification: Given poor performance in risk-focused SVM models, as well as great overlap of risk factors across individuals with and without Alzheimer's, post-hoc analyses with both risk factors and symptomology were conducted to investigate changes to model performance. The linear SVM was applied to the full dataset without prior LLE data reduction. As shown in Table 12, this model received an accuracy of 81.71%, indicating substantially improved classification of our sample. Class-wide results showed that the model classified healthy individuals, marked as class 0 (F1-score = 0.85), quite well compared to

Alzheimer's patients, marked as class 1 (F1-score = 0.76). This suggests a slight bias towards correct classification of the majority class of healthy individuals, but improved overall classification. The confusion matrix (Figure 19) showed that the model produces substantially fewer misclassifications for both healthy individuals and Alzheimer's patients, visualized by the decreased number of false positives and false negatives. Moreover, the model showed marked improvement in detecting individuals with Alzheimer's disease (recall = 0.83). Lastly, the ROC curve (Figure 20; ROC AUC = 89.08%) has significant area under the curve, indicating good overall model performance.

Post-hoc Broad classification: No LLE and SVM RBF kernel. The non-linear RBF SVM was applied to the full dataset without prior LLE data reduction. As shown in Table 14, this model received an accuracy of 82.95%, indicating good classification of our sample. Class-wide results showed that the model classified healthy individuals, marked as class 0 (F1-score = 0.86), and Alzheimer's patients, marked as class 1 (F1-score = 0.77) substantially well. This suggests a slight bias towards correct classification of the majority class of healthy individuals, but improved overall classification. The confusion matrix (Figure 21) showed that the model produces substantially fewer misclassifications for both healthy individuals and Alzheimer's patients, visualized by the decreased number of false positives and false negatives. Moreover, the model showed comparable performance in detecting individuals with Alzheimer's disease (recall = 0.82). Lastly, the ROC curve (Figure 22; ROC AUC = 89.92%) has significant area under the curve, confirming good overall model performance.

Post-hoc Broad classification: LLE and SVM linear kernel. The linear SVM was applied to the full dataset with prior LLE data reduction. As shown in Table 13, this model received an accuracy of 62.64%, indicating moderate classification of our sample. Class-wide

results showed that the model classified healthy individuals, marked as class 0 (F1-score = 0.71), substantially better than Alzheimer's patients, marked as class 1 (F1-score = 0.48). This suggests a bias towards correct classification of the majority class of healthy individuals. The confusion matrix (Figure 23) showed that the model produces an abundance of misclassifications for both healthy individuals and Alzheimer's patients, visualized by the large number of false positives and false negatives. Moreover, the model misclassifies patients with Alzheimer's more often than it correctly classifies them. The low recall for Alzheimer's patients (0.48) relative to healthy individuals (0.71), suggests that the model performs poorer in detecting individuals with Alzheimer's disease. Lastly, the ROC curve (Figure 24; ROC AUC = 62.10%) approximately follows the diagonal line, with some increased area under the area, confirming moderate overall model performance.

Post-hoc Broad classification: LLE and SVM RBF kernel. The non-linear RBF SVM was applied to the full dataset with prior LLE data reduction. As shown in Table 15, this model received an accuracy of 62.64%, indicating moderate classification of our sample. Class-wide results showed that the model classified healthy individuals, marked as class 0 (F1-score = 0.72), substantially better than Alzheimer's patients, marked as class 1 (F1-score = 0.43). This suggests a bias towards correct classification of the majority class of healthy individuals. The confusion matrix (Figure 25) showed that the model produces an abundance of misclassifications for both healthy individuals and Alzheimer's patients, visualized by the large number of false positives and false negatives. Moreover, the model misclassifies patients with Alzheimer's more often than it correctly classifies them. The low recall for Alzheimer's patients (0.40) relative to healthy individuals (0.75), suggests that the model performs poorer in detecting individuals with Alzheimer's disease. Lastly, the ROC curve (Figure 26; ROC AUC = 62.23%) approximately

follows the trend of the diagonal line, often with increased area under the curve. This further confirms our prior findings of moderate overall model performance.

Interpretation & Future Works

Summary Results

Through this project, we aimed to classify Alzheimer's patients from healthy individuals by leveraging data reduction and classification algorithms prevalent in machine learning implementations. We examined whether reducing the dataset using LLE aided downstream classification algorithms. Moreover, we investigated whether leveraging different classification methods, namely Decision-Tree or Support Vector Machine, resulted in varied ability to classify our sample. We also conducted post-hoc analyses, investigating whether the inclusion of symptom-focused variables can improve model performance.

Overall, we found that model performance was highest, at an ROC AUC of 93.76%, when leveraging the full dataset dimensionality reduction and classifying through the DT algorithm. This was followed closely by the SVM algorithm, in which the full dataset without dimensionality reduction was leveraged with the non-linear RBF kernel, resulting in an ROC accuracy of 82.95%. Across the DT algorithms without dimension reduction, "Functional Assessment" best classified our sample when utilizing the full dataset, while "Sleep Quality" best classified our sample when utilizing the risk-focused dataset. Sleep Quality may reflect a key marker for detecting Alzheimer's disease and should be further investigated for its utility in clinical screenings. It is also possible that risk factors outside of the scope of the present analyses may be more suitable for clinical screenings. Moreover, it is important to note that Functional Assessment is closely tied to Alzheimer's diagnosis and may reflect disease presence, rendering it inappropriate for risk detection purposes. Variables that best classified our sample in

DT algorithms with dimension reduction were not analyzed because the resultant dimensions of LLE are not interpretable, nor clinically meaningful. Within the SVM implementation, classifying our sample with either the linear or non-linear RBF kernel resulted in similar model performance metrics. This suggests that linear combinations of the variables adequately classify our sample.

Across the DT and SVM algorithms, a general pattern emerged: utilizing the full dataset without prior dimensionality reduction aided model performance. This suggests that, while leveraging the full dataset negates our primary objective to produce a risk-focused classifier, integrating cognitive, behavioural and other symptom-focused variables aids classification efforts. Moreover, dimensionality reduction hindered model performance, particularly for the SVM algorithm. An explanation may be that LLE is an unsupervised algorithm, which preserves the local structure of data without considering whether those data points represent individuals with or without Alzheimer's disease. Thus, this prior data reduction may distort our data, particularly when honing in on the risk-focused dataset, which has substantial overlap of individuals with and without Alzheimer's disease in any given risk factor. Given that SVM is well-suited towards classifying data with high dimensionality (Cortes & Vapnik, 1995), we conclude that prior dimensionality reduction through LLE hinders optimization of the SVM algorithm.

Were these results as expected?

These results were partly aligned with our hypotheses. Prior to our analyses, we did not have a hypothesis regarding which classifier (Decision Tree or Support Vector Machine) would perform better. We had hypothesized that cardiovascular disease or alcohol consumption in the risk-focused dataset would best classify our sample, as these are known to be impactful risk

factors for dementia-related diseases (Livingston et al., 2020). Moreover, we hypothesized that Mini-Mental State Examination (MMSE) scores in the full dataset would best classify our sample, as this captures cognitive decline across many domains and is often used to assess individuals for Alzheimer's disease (Folstein et al.,1975)). However, neither cardiovascular disease, alcohol consumption, nor MMSE scores best classified our sample. We correctly hypothesized that models trained on the full dataset would achieve higher classification accuracy due to the close links between symptom-focused variables and Alzheimer's disease. Moreover, we expected dimensionality reduction using LLE to improve model performance, given the relatively high number of variables in the dataset. However, LLE consistently hindered model performance across both DT and SVM models. Lastly, we expected that the non-linear RBF kernel would outperform the linear kernel in SVM due to its greater flexibility in capturing complex relationships between variables. However, both kernels produced similar results across conditions.

Limitations and next steps

Within the implementation of our LLE data reduction, we did not optimize the parameters of dimensions and neighbours to maximize model performance. Instead, fixed values of 5 dimensions and 15 neighbours were chosen to ensure sufficient retention of information for subsequent classification. The lack of optimization of the LLE data reduction could explain why models that included data reduction were performing near chance. Future studies should optimize these parameters. Moreover, only two kernels were evaluated within the implementation of SVM. While the linear and RBF kernels represent the most basic and popular boundaries, future research should investigate other kernels, such as polynomial and sigmoid, which may yield boundaries that better classify our sample. Lastly, while models trained on the

full dataset performed substantially well in classifying individuals with and without Alzheimer's, these models are not clinically meaningful for risk-detection purposes. Thus, future research should expand the present analyses by honing in on other risk factors, such as hearing loss, isolation, genetics, etc. Models trained on an expanded list of risk factors may better classify individuals with and without Alzheimer's disease.

Distribution of Tasks

Anusha completed the introduction portion. Zoe completed the data and preprocessing sections and the exploratory data analyses portion. The method and results portion discussing the DT algorithms were completed by Zoe. The method and results portions discussing the LLE and SVM algorithms were completed by Khushi. The discussion was completed by Khushi. The code was completed in part by Anusha, Zoe and Khushi. Anusha coded the LLE algorithm, and split the datasets across both training and test sets, as well as risk-only and the full dataset. Zoe visualized the data descriptors and coded exploratory data analyses, as well as the Decision Tree algorithms. Khushi coded the SVM algorithms and completed the read.me file.

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Tables

Table 1. Variables

Variables (32)	Demographics (categorical): PatientID, Age, Gender, Ethnicity, EducationLevel
	Lifestyle Factors: BMI, Smoking, AlcoholConsumption, PhysicalActivity, DietQuality, SleepQuality
	Medical History (categorical): FamilyHistoryAlzheimers, CardiovascularDisease, Diabetes, Depression, HeadInjury, Hypertension
	Clinical Measurements: SystolicBP, DiastolicBP, CholesterolTotal, CholesterolLDL, CholesterolHDL, CholesterolTriglycerides
	Cognitive & Functional Assessments: Mini-Mental State Examination score (MMSE), FunctionalAssessment, MemoryComplaints, BehavioralProblems, Activities of Daily Living Score (ADL)
	Symptoms: Confusion, Disorientation, PersonalityChanges, DifficultyCompletingTasks, Forgetfulness
Outcome	Diagnosis (binary, 0 = healthy, 1 = Alzheimer's)

Table 2. Demographics

Variable	Summary
Sample size	2,149 (Alzheimer's: 759 = 35.4%, Healthy: 1,390 = 64.6%)
Age	75 ± 9 years
Gender	50% Female, 50% Male
Ethnicity	60% Caucasian, 20% African American, 10% Asian, 10% Other

Education Level	20.8% None, 39.7% High School, 29.6% Bachelor's, 9.9% Higher

Table 3. Continuous Variables Summary

Variable	Mean	SD	Min	Max
Age	74.91	8.99	60.0	90.0
BMI	27.66	7.22	15.01	39.99
AlcoholConsumption	10.04	5.76	0.00	19.99
PhysicalActivity	4.92	2.86	0.00	9.99
DietQuality	4.99	2.91	0.01	10.00
SleepQuality	7.05	1.76	4.00	10.00
SystolicBP	134.26	25.95	90.0	179.0
DiastolicBP	89.85	17.59	60.0	119.0
CholesterolTotal	225.20	42.54	150.09	299.99
CholesterolLDL	124.34	43.37	50.23	199.97
CholesterolHDL	59.46	23.14	20.00	99.98
CholesterolTriglycerides	228.28	101.99	50.41	399.94
MMSE	14.76	8.61	0.01	29.99
FunctionalAssessment	5.08	2.89	0.00	10.00
ADL	4.98	2.95	0.00	10.00

Table 4. Top 5 Variables Correlated With Diagnosis

Variable	Correlation	Interpretation

FunctionalAssessment	−0.365	Lower daily function → more likely Alzheimer's
ADL	−0.332	Lower daily living ability → more likely Alzheimer's
MemoryComplaints	+0.307	Presence of complaints → more likely Alzheimer's
MMSE	−0.237	Lower cognitive score → more likely Alzheimer's
BehavioralProblems	+0.224	Presence of problems → more likely Alzheimer's

Table 5. Performance Metrics for the Decision Tree Classifier on the Risk Dataset Without LLE.

	Precision	Recall	F1-score	Support
Healthy Individuals	0.65	0.56	0.60	417
Alzheimer's patients	0.36	0.46	0.40	228
Accuracy			0.52	645
Macro Average	0.51	0.51	0.50	645
Weighted Average	0.55	0.52	0.53	645

Table 6. Performance Metrics for the Decision Tree Classifier on the Risk Dataset With LLE.

	Precision	Recall	F1-score	Support

Healthy Individuals	0.64	0.46	0.53	417
Alzheimer's patients	0.35	0.53	0.42	228
Accuracy			0.48	645
Macro Average	0.49	0.49	0.48	645
Weighted Average	0.54	0.48	0.49	645

Table 7. Performance Metrics for the Decision Tree Classifier on the Full Dataset Without LLE.

	Precision	Recall	F1-score	Support
Healthy Individuals	0.95	0.95	0.95	417
Alzheimer's patients	0.91	0.91	0.91	228
Accuracy			0.94	645
Macro Average	0.93	0.93	0.93	645
Weighted Average	0.94	0.94	0.94	645

Table 8. Performance Metrics for the Decision Tree Classifier on the Full Dataset With LLE.

	Precision	Recall	F1-score	Support

Healthy Individuals	0.66	0.97	0.79	417
Alzheimer's patients	0.61	0.09	0.15	228
Accuracy			0.66	645
Macro Average	0.63	0.53	0.47	645
Weighted Average	0.64	0.66	0.56	645

Table 9 *Performance Metrics for the Linear SVM on the Risk Dataset Without LLE*

	Precision	**Recall**	**F1 score**	**Support**
Healthy Individuals	0.66	0.49	0.56	417
Alzheimer's patients	0.36	0.54	0.43	228
Accuracy			0.51	645
Macro Average	0.51	0.51	0.50	645
Weighted Average	0.55	0.51	0.52	645

Table 10 Performance Metrics for the Linear SVM on the Risk Dataset With LLE

	Precision	**Recall**	**F1 score**	**Support**
Healthy Individuals	0.65	1.00	0.79	417

Alzheimer's patients	0.00	0.00	0.00	228
Accuracy			0.65	645
Macro Average	0.32	0.50	0.39	645
Weighted Average	0.42	0.65	0.51	645

Table 11 Performance Metrics for the SVM RBF Kernel on the Risk Dataset With LLE

	Precision	**Recall**	**F1 score**	**Support**
Healthy Individuals	0.62	0.57	0.60	417
Alzheimer's patients	0.32	0.37	0.34	228
Accuracy			0.50	645
Macro Average	0.47	0.47	0.47	645
Weighted Average	0.52	0.50	0.51	645

Table 12 Performance Metrics for linear SVM on Full Dataset Without LLE

	Precision	**Recall**	**F1 score**	**Support**
Healthy Individuals	0.90	0.81	0.85	417
Alzheimer's patients	0.70	0.83	0.76	228

Accuracy			0.82	645
Macro Average	0.80	0.82	0.81	645
Weighted Average	0.83	0.82	0.82	645

Table 13 Performance Metrics for Linear SVM on Full Dataset With LLE

	Precision	**Recall**	**F1 score**	**Support**
Healthy Individuals	0.71	0.71	0.71	417
Alzheimer's patients	0.47	0.48	0.48	228
Accuracy			0.63	645
Macro Average	0.59	0.59	0.59	645
Weighted Average	0.63	0.63	0.63	645

Table 14 Performance Metrics for SVM RBF kernel on Full Dataset Without LLE

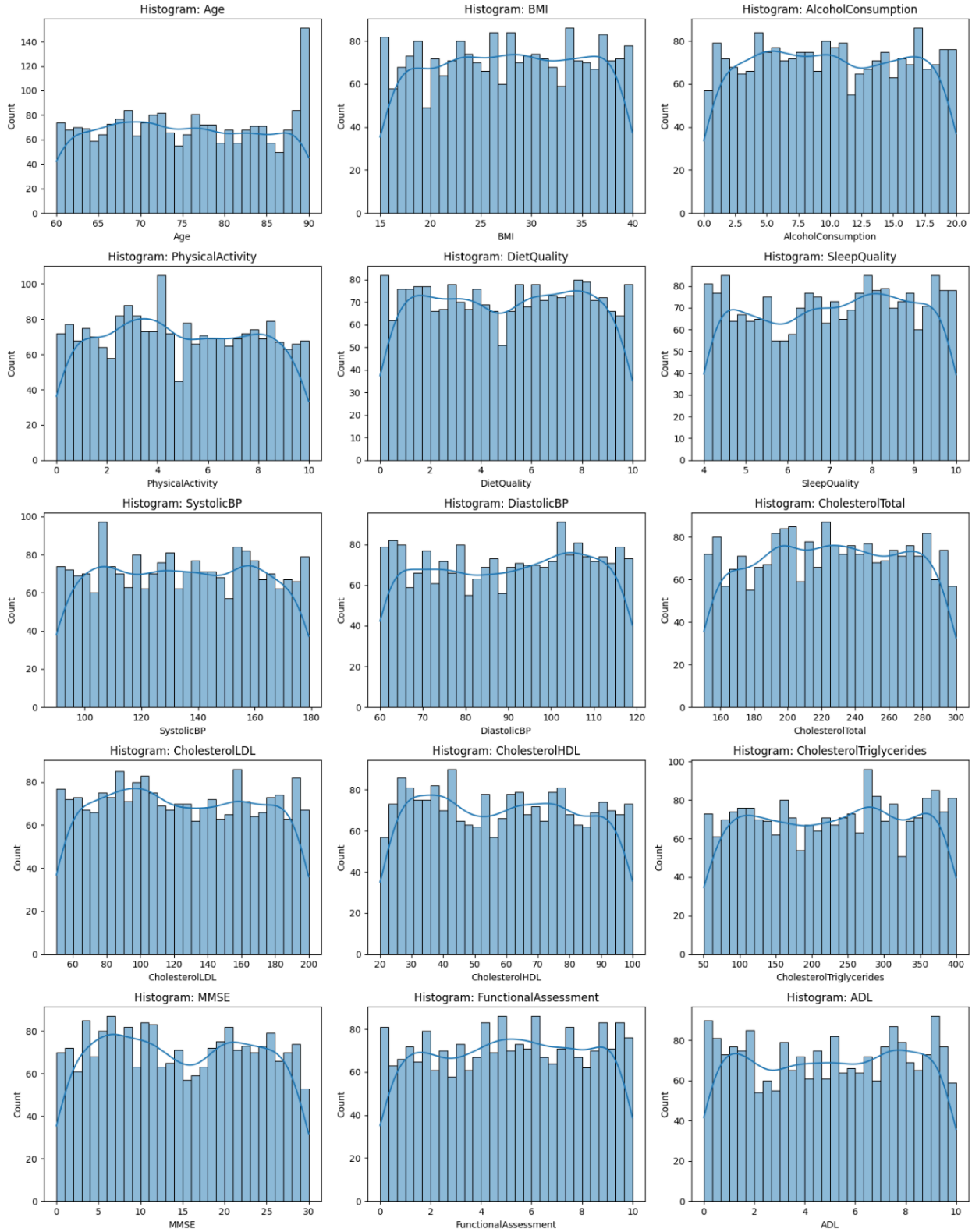
	Precision	**Recall**	**F1 score**	**Support**
Healthy Individuals	0.90	0.83	0.86	417
Alzheimer's patients	0.73	0.82	0.77	228
Accuracy			0.83	645

Macro Average	0.81	0.83	0.82	645
Weighted Average	0.84	0.83	0.83	645

Table 15 Performance Metrics for SVM RBF kernel on Full Dataset With LLE

	Precision	**Recall**	**F1 score**	**Support**
Healthy Individuals	0.70	0.75	0.72	417
Alzheimer's patients	0.47	0.40	0.43	228
Accuracy			0.63	645
Macro Average	0.58	0.58	0.58	645
Weighted Average	0.62	0.63	0.62	645

Figures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SleepQuality
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SystolicBP
Histogram: DiastolicBP
Count
DiastolicBP
Histogram: CholesterolTotal
Count
CholesterolTotal
Histogram: CholesterolLDL
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CholesterolLDL
Histogram: CholesterolHDL
Count
CholesterolHDL
Histogram: CholesterolTriglycerides
Count
CholesterolTriglycerides
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FunctionalAssessment
Histogram: ADL
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Figure 1. Representative histograms of continuous variables.

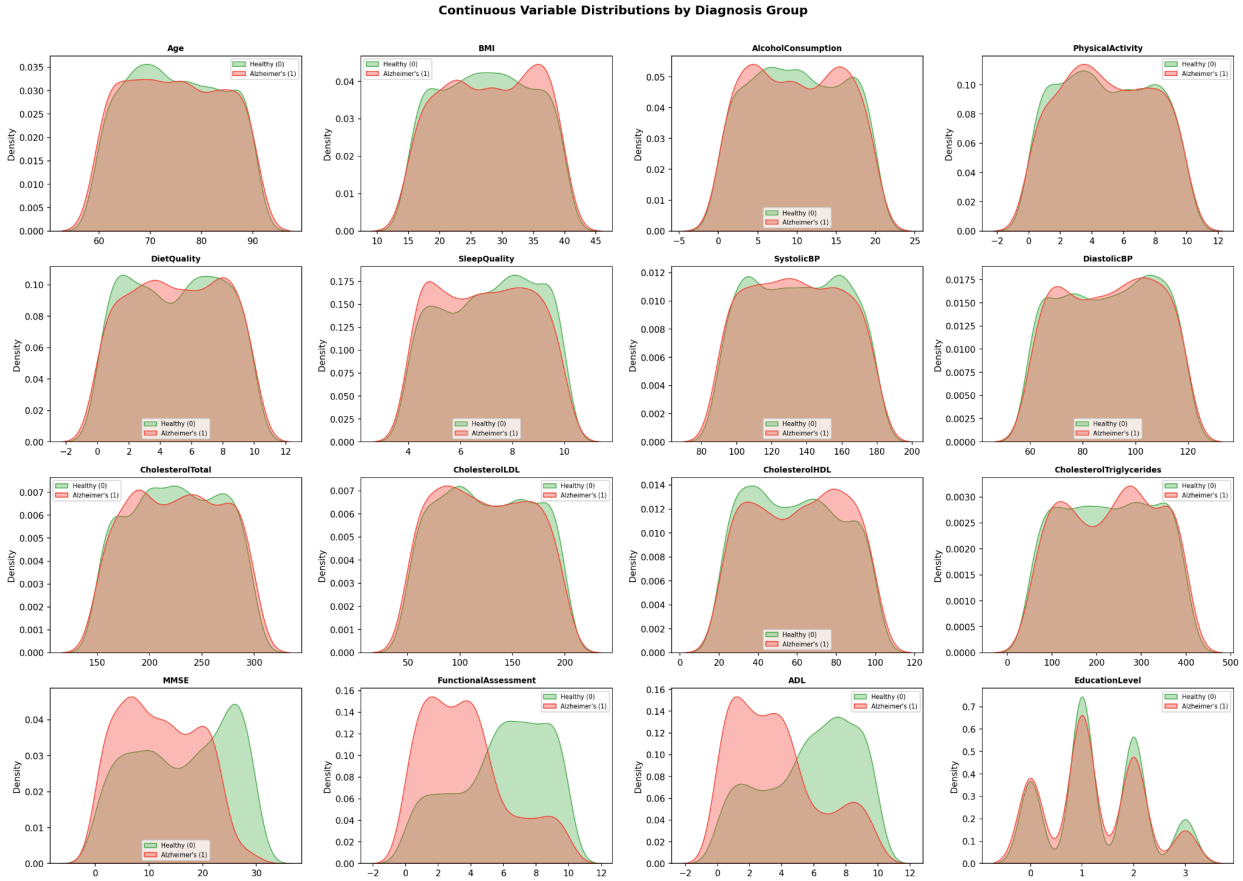

Figure 2. KDE overlay of continuous variables by diagnosis group. MMSE, Functional Assessment, and ADL show the clearest separation.

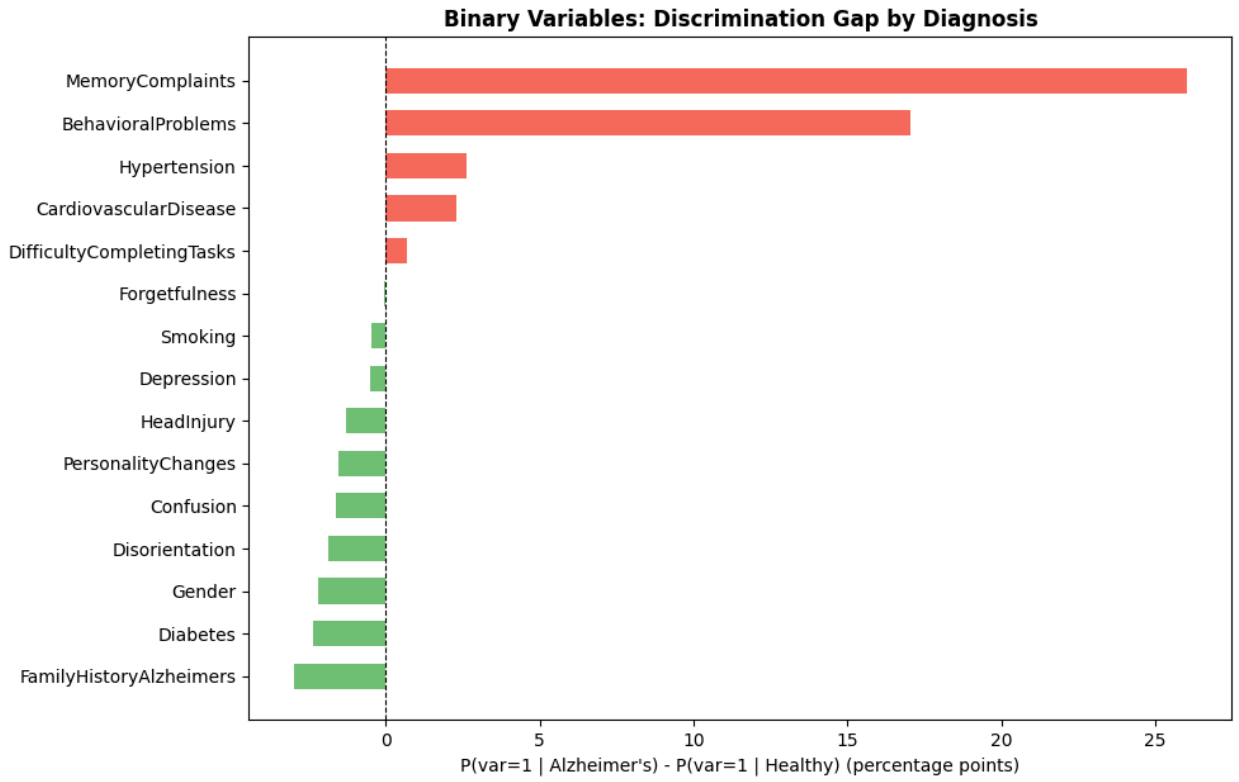

Figure 3. Discrimination gap for binary variables: P(var=1|Alzheimer's) − P(var=1|Healthy).

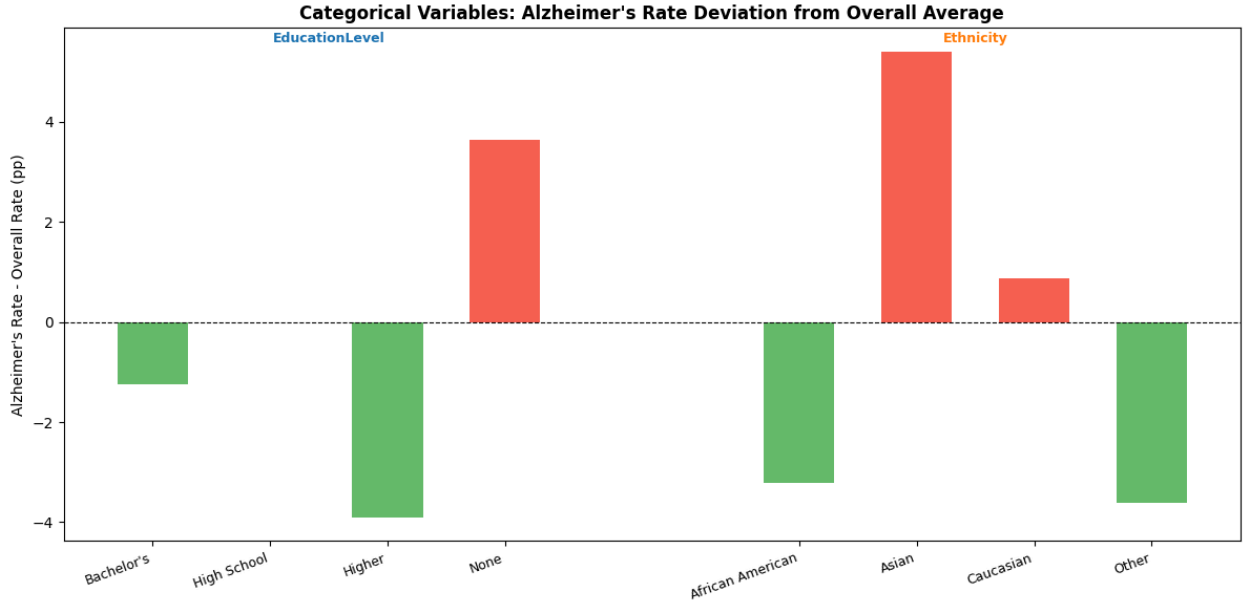

Figure 4. Alzheimer's rate deviation from the overall average (35.4%) by category level.

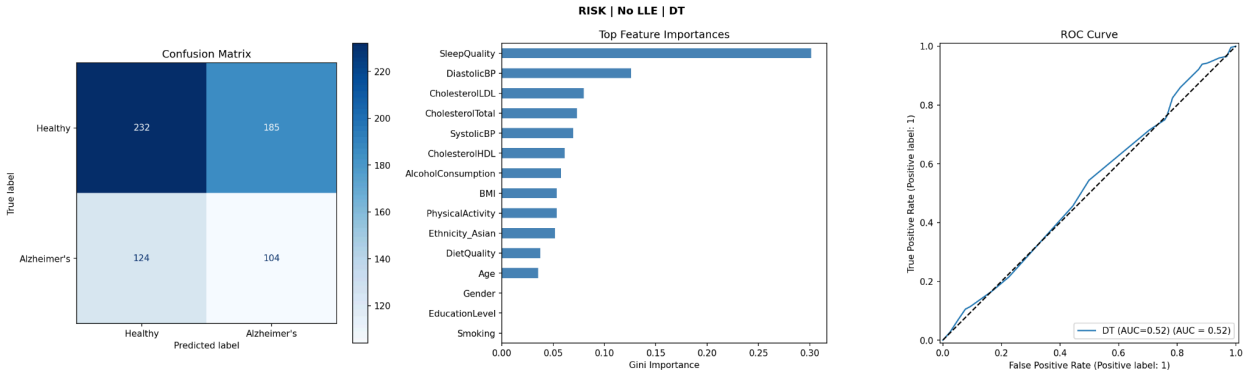

Figure 5. Risk-only, No LLE + DT: confusion matrix, feature importances, and ROC curve.

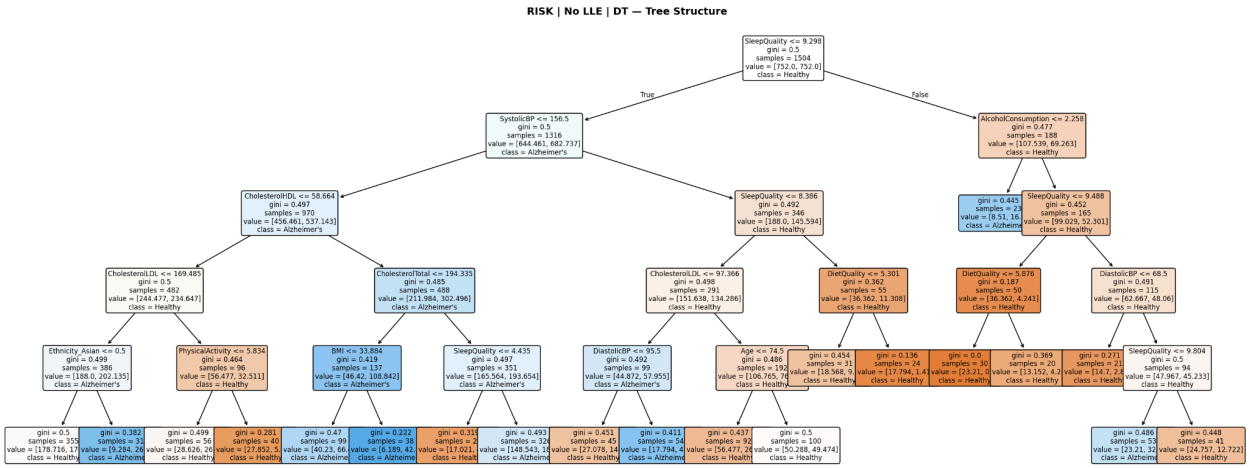

Figure 6. Decision tree structure — Risk-only, No LLE.

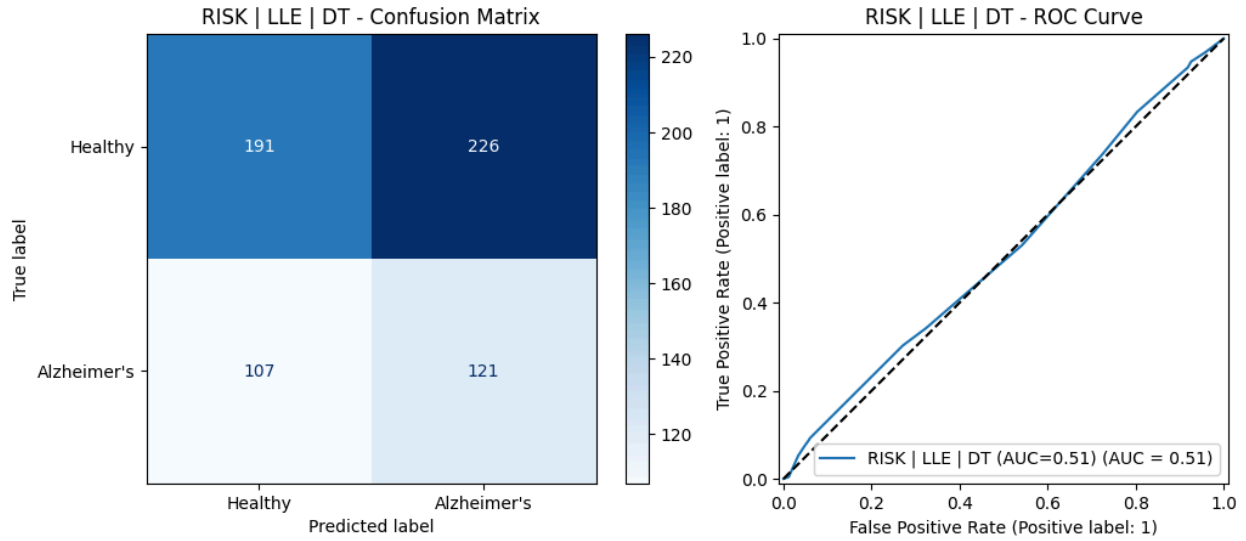

Figure 7. Risk-only, LLE + DT: confusion matrix and ROC curve.

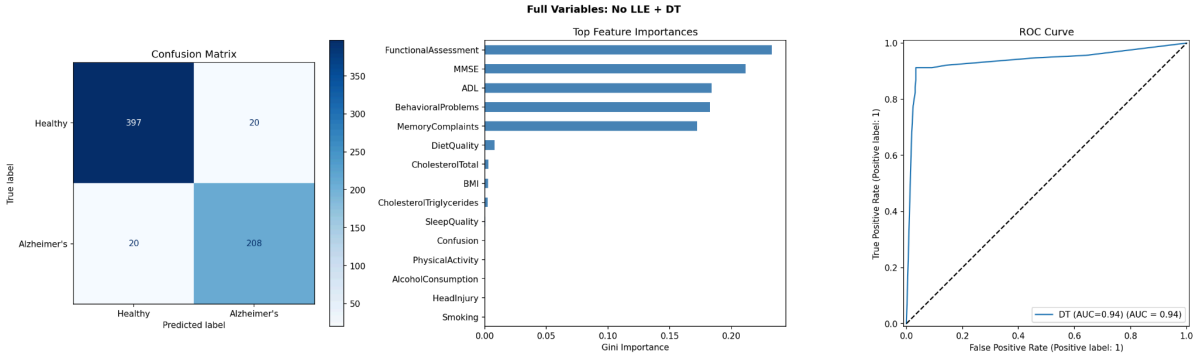

Figure 8. Full variables, No LLE + DT: confusion matrix, feature importances, and ROC curve.

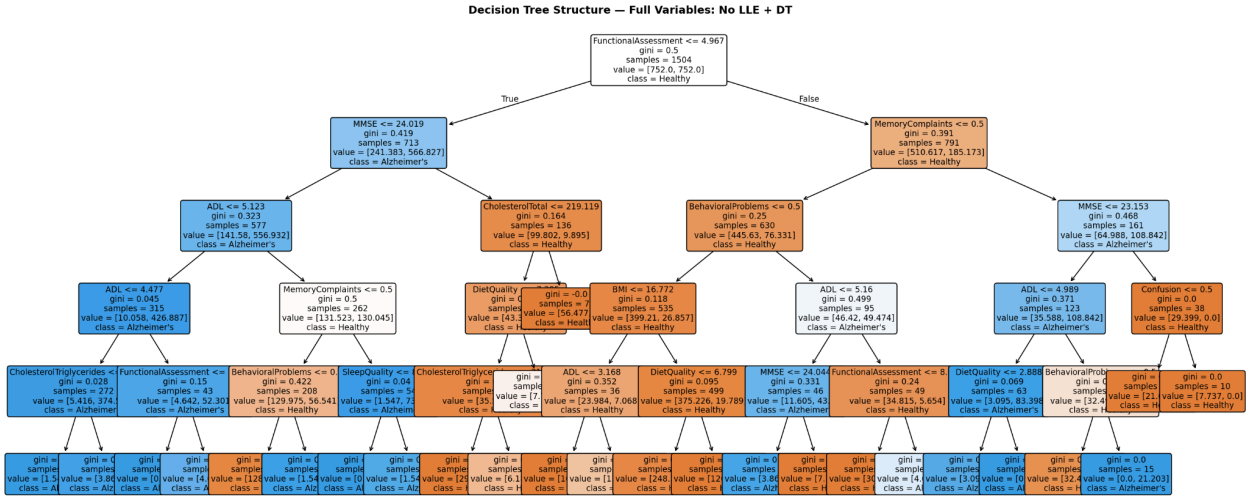

Figure 9. Decision tree structure — Full variables, No LLE.

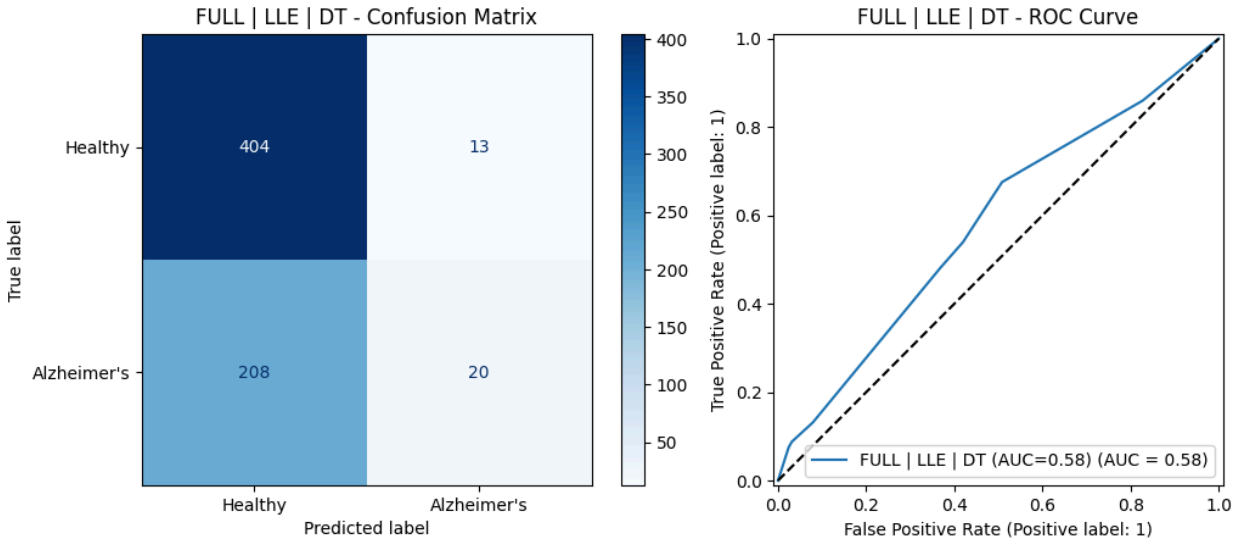

Figure 10. Full variables, LLE + DT: confusion matrix and ROC curve.

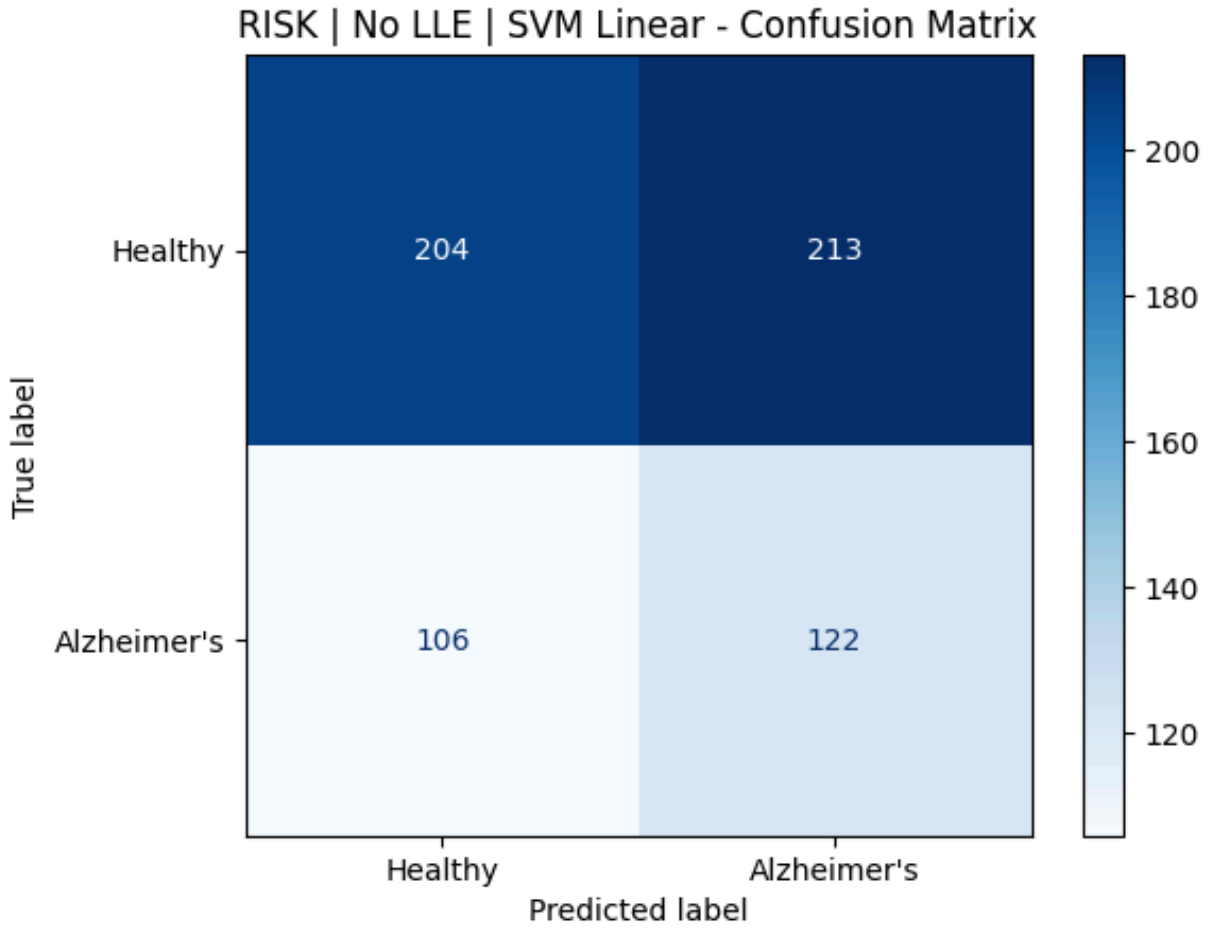

Figure 11. Risk-only, no LLE + SVM linear kernel: confusion matrix.

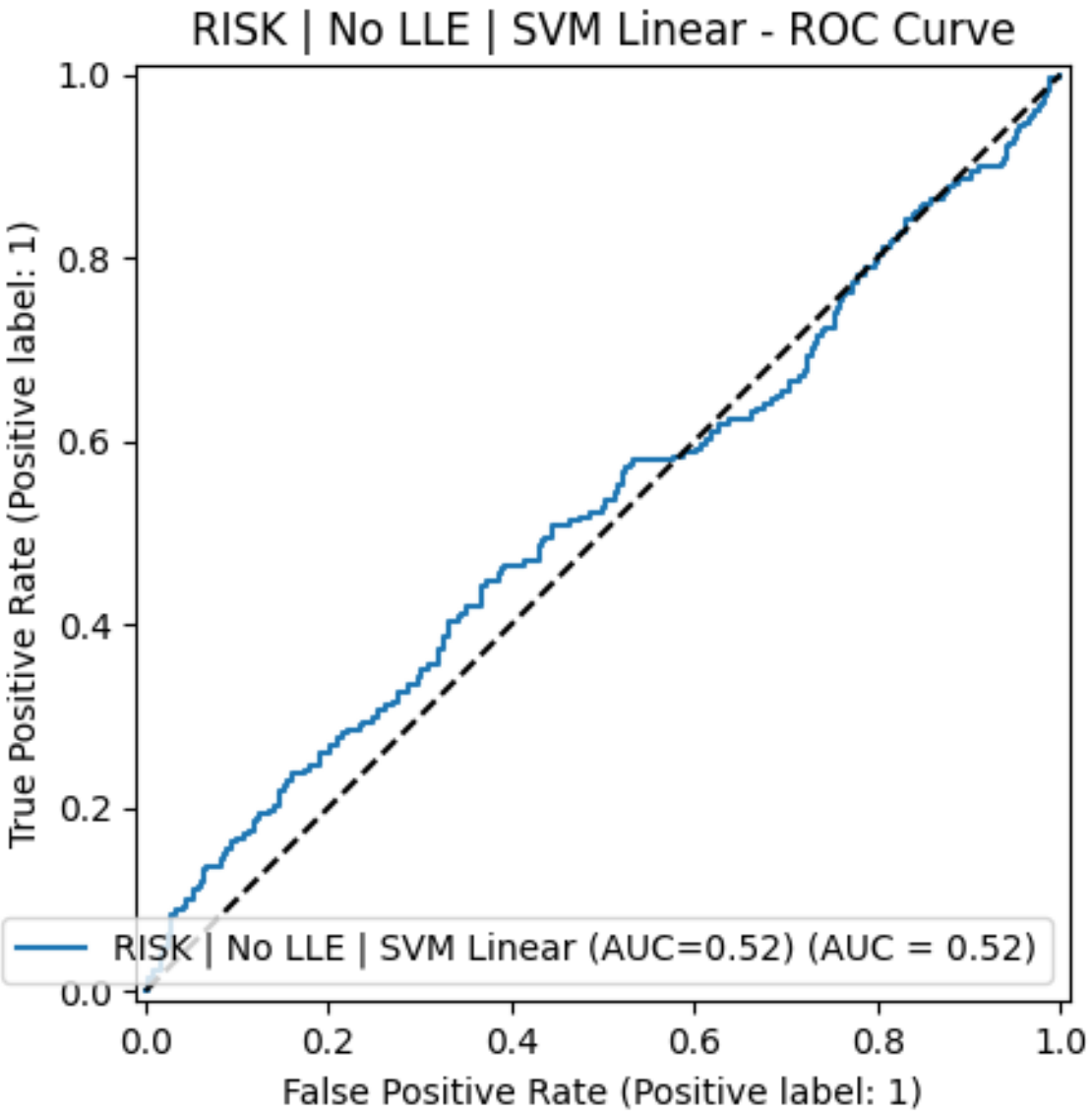

Figure 12. Risk-only, no LLE + SVM linear kernel: ROC AUC curve,.

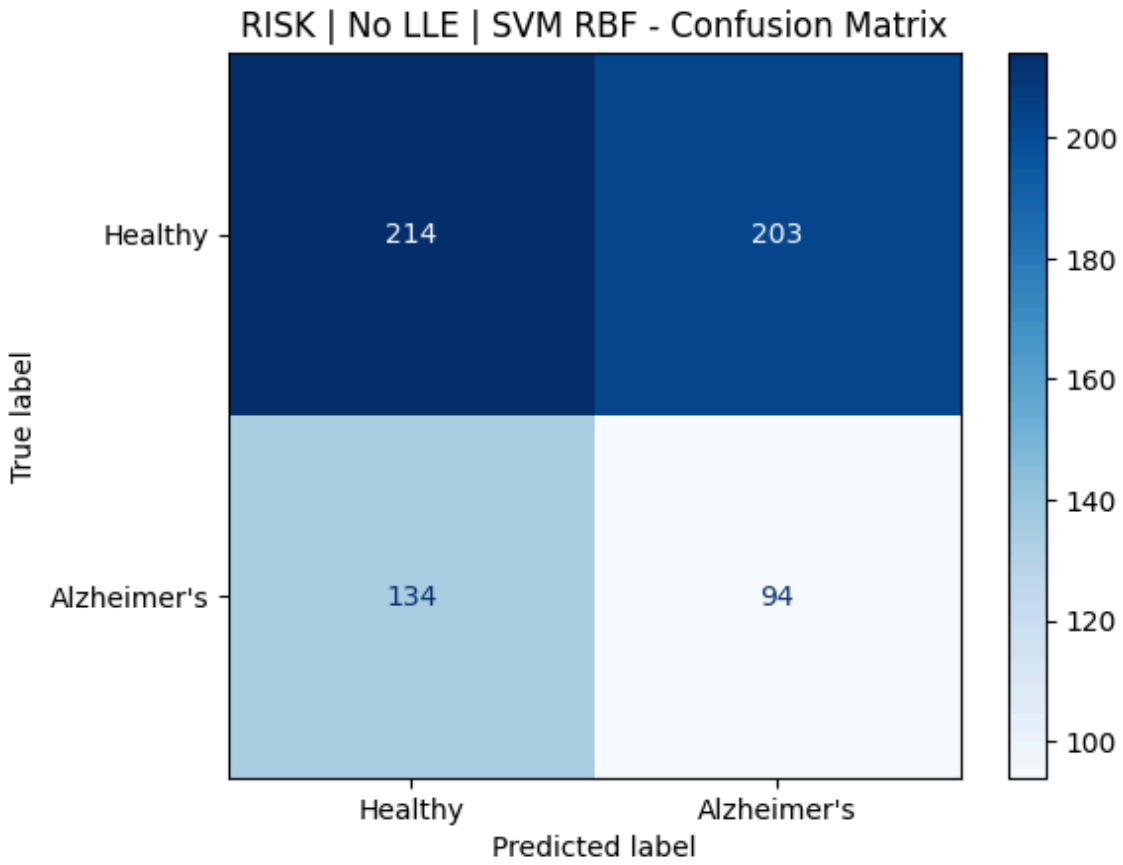

Figure 13. Risk-only, no LLE + SVM RBF kernel: confusion matrix.

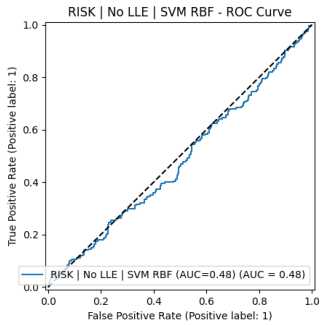

Figure 14. Risk-only, no LLE + SVM RBF kernel: ROC AUC curve.

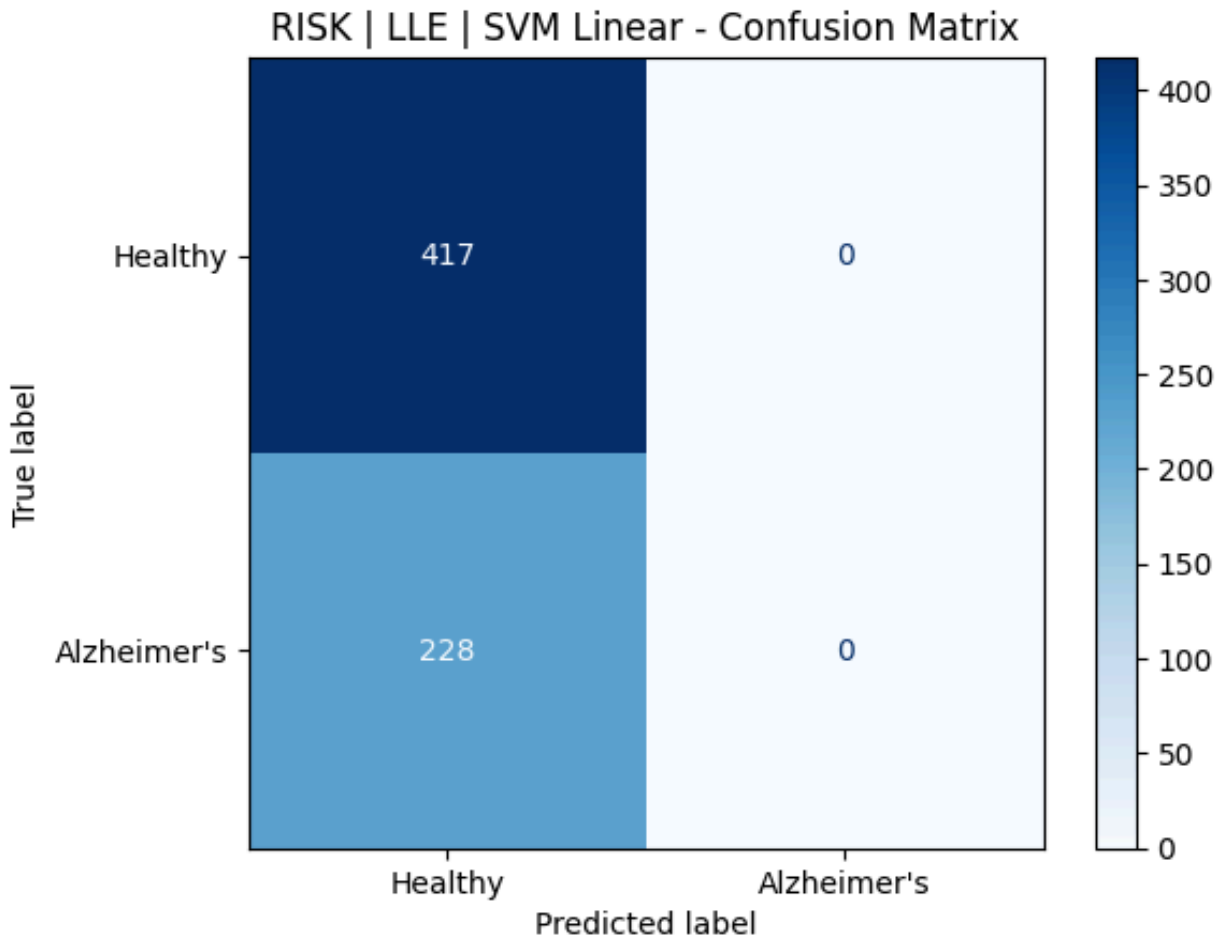

Figure 15. Risk-only, LLE + SVM linear kernel: confusion matrix.

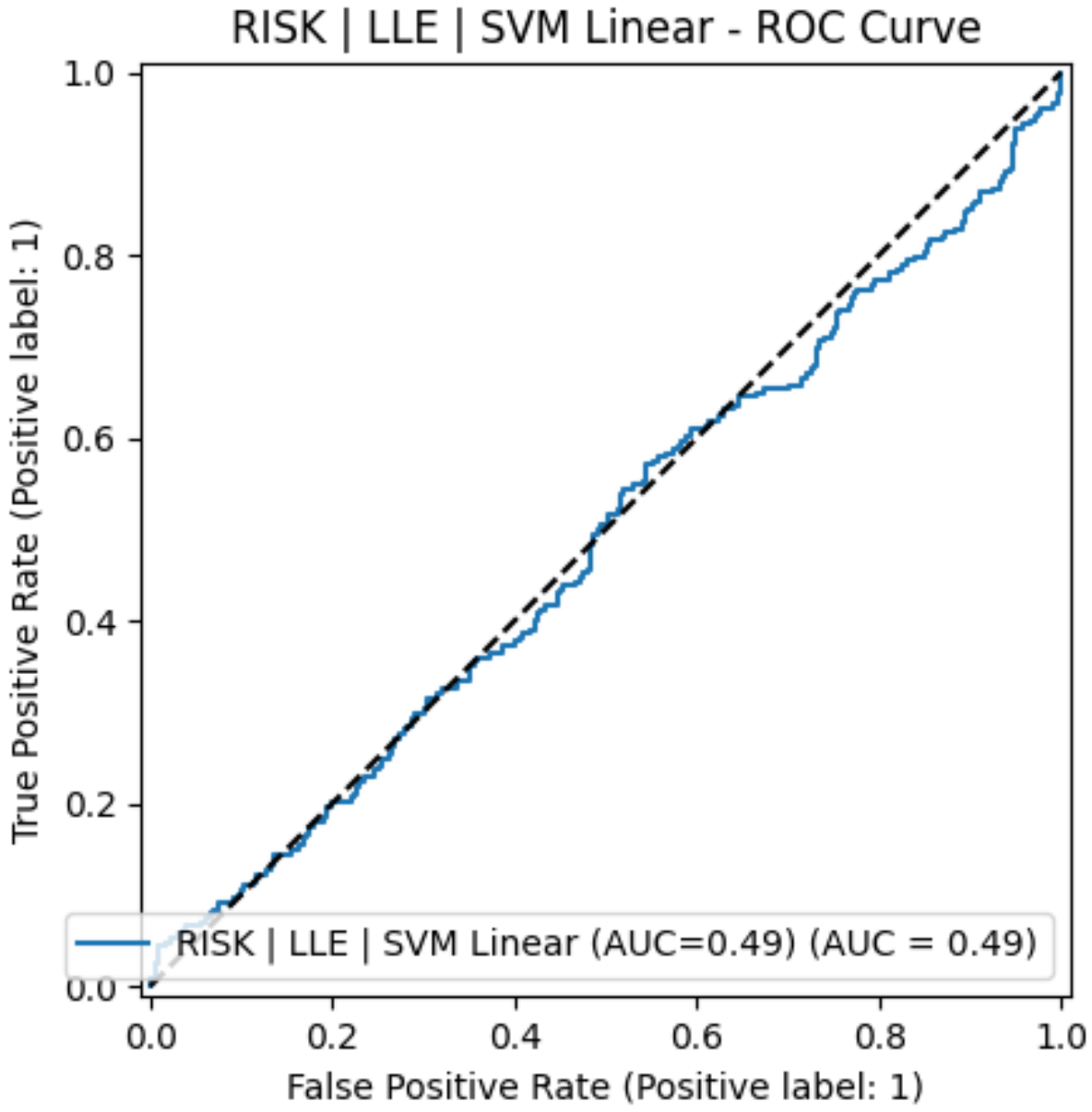

Figure 16. Risk-only, LLE + SVM RBF kernel: ROC AUC curve.

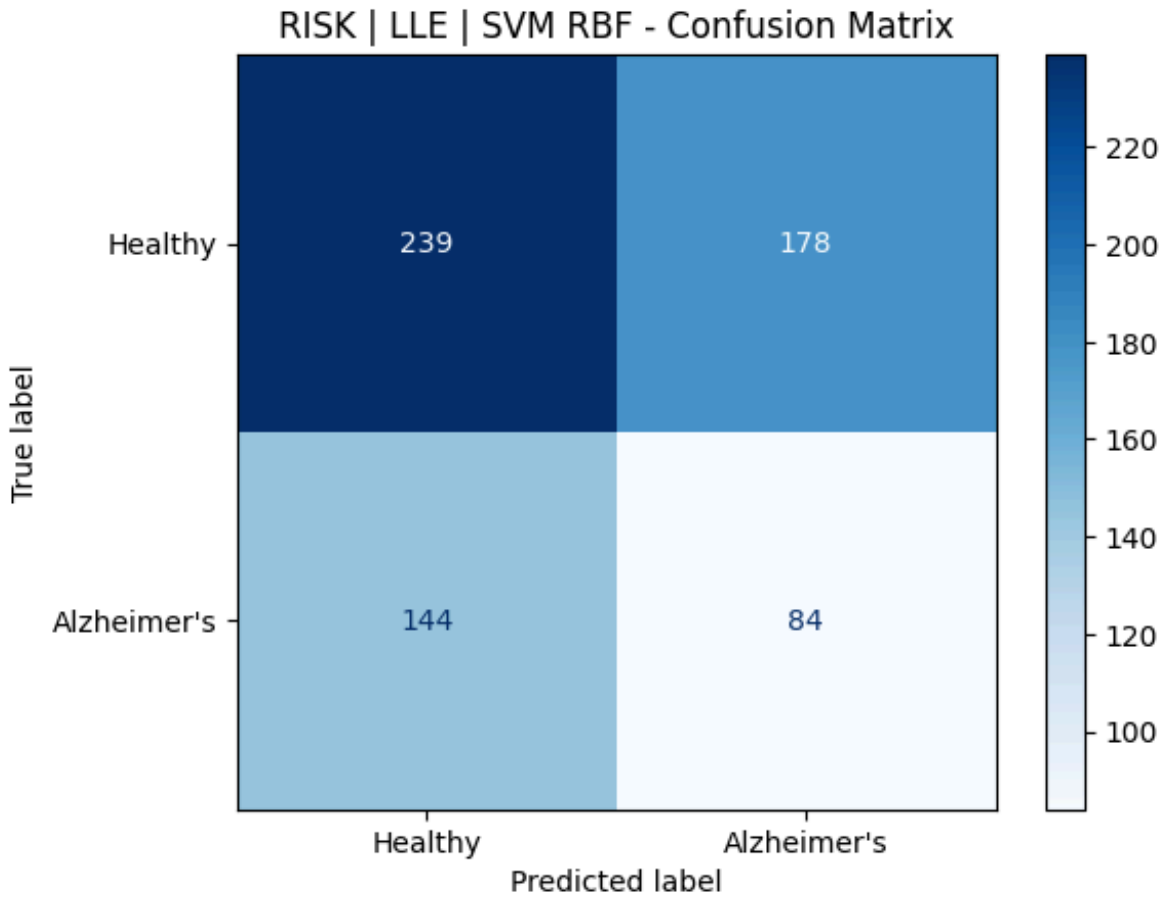

Figure 17. Risk-only, LLE + SVM RBF kernel: confusion matrix.

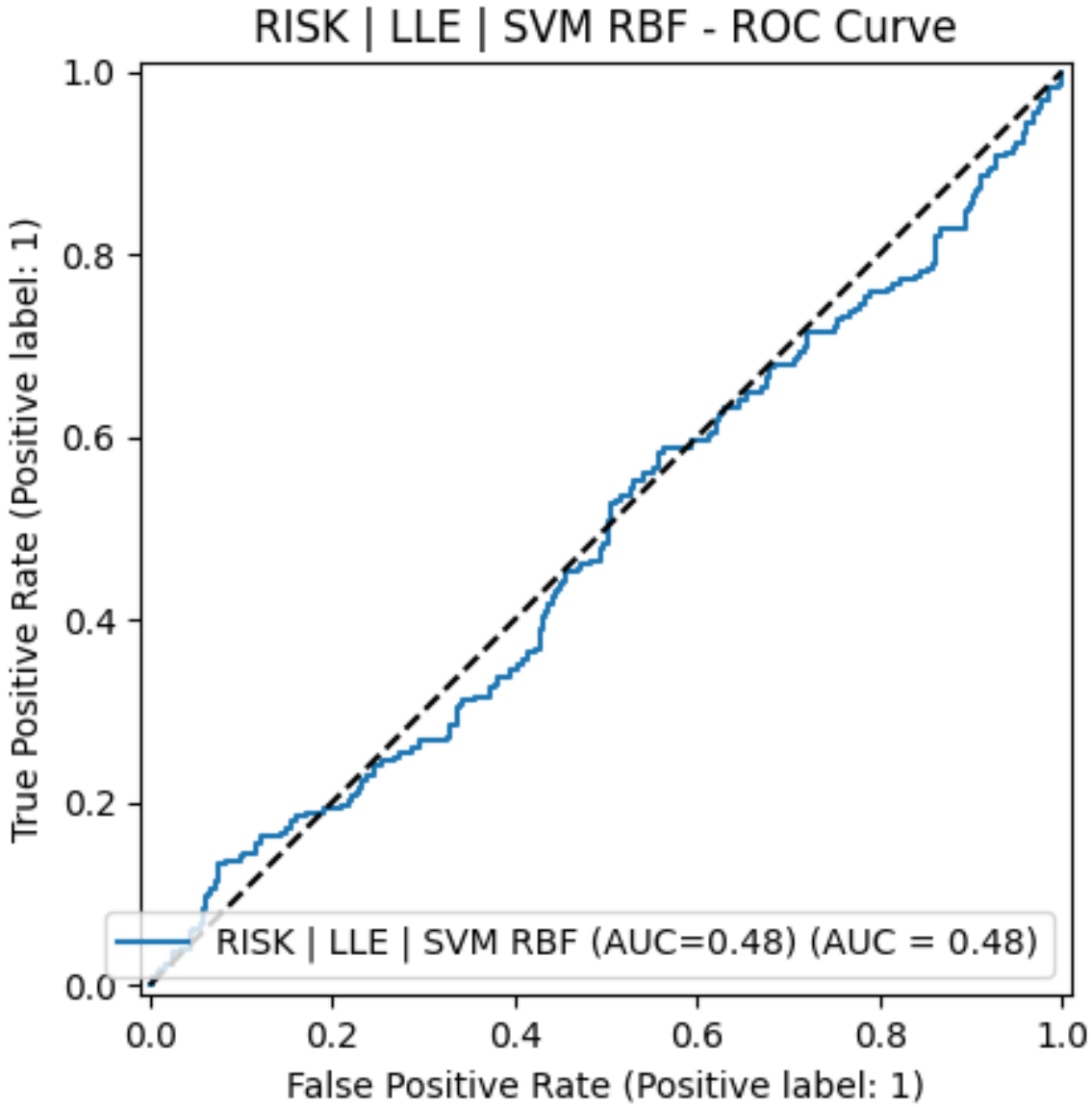

Figure 18. Risk-only, LLE + SVM RBF kernel: ROC AUC curve.

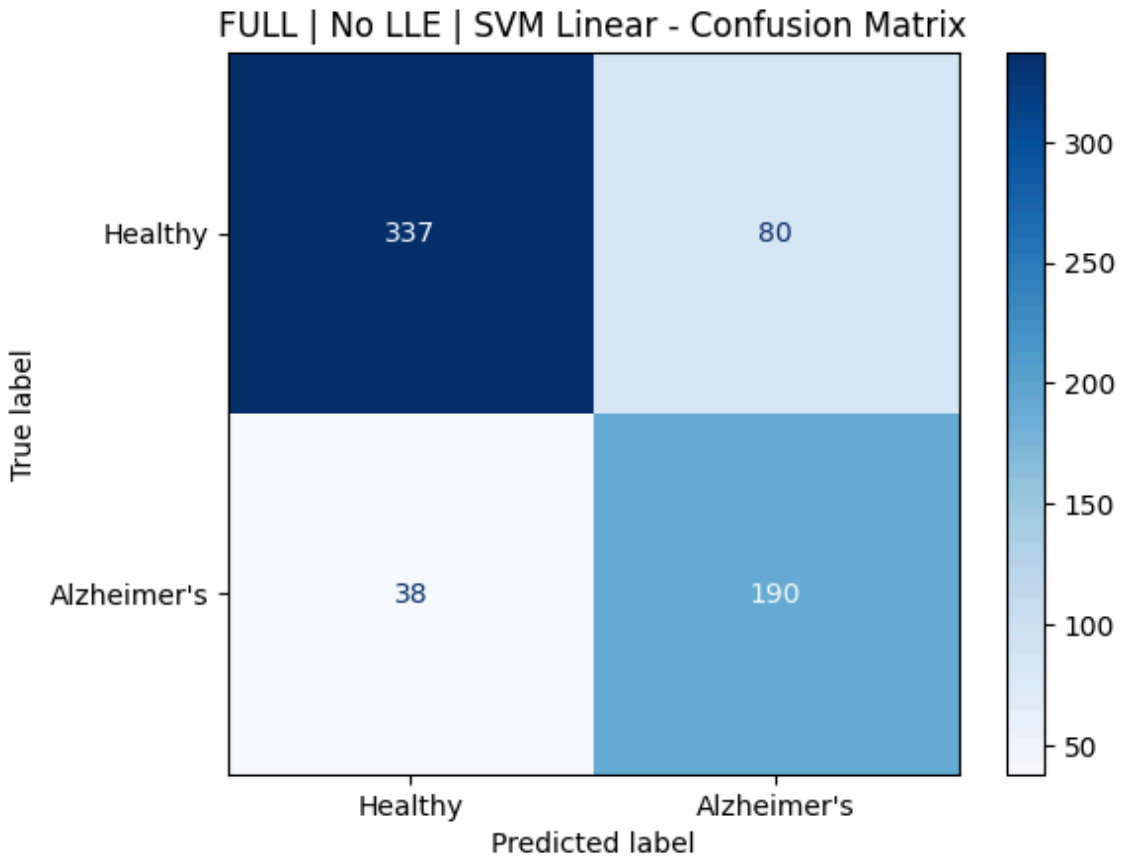

Figure 19. Full variables, no LLE + SVM linear kernel: confusion matrix.

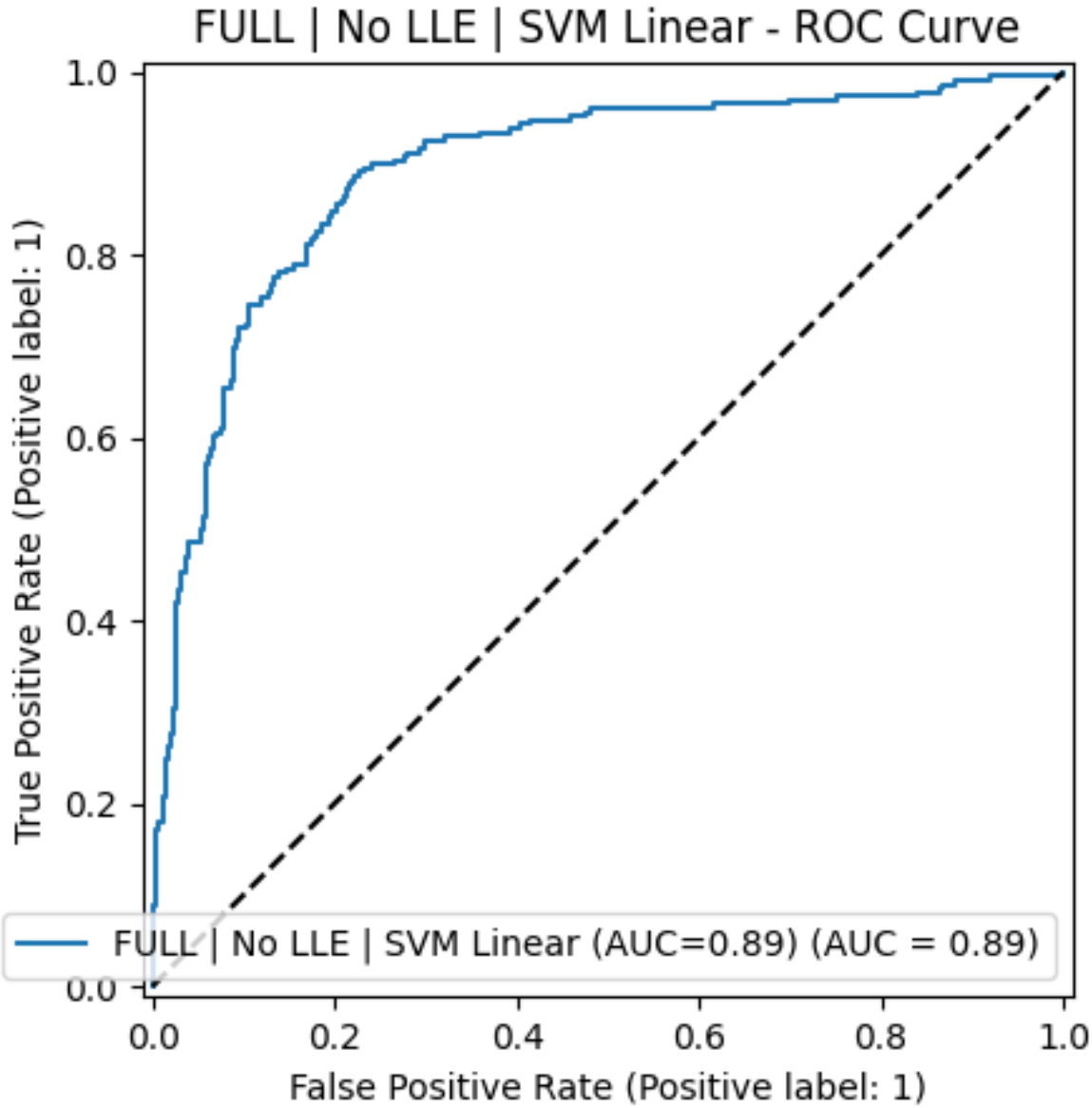

Figure 20. Full variables, no LLE + SVM linear kernel: ROC AUC curve.

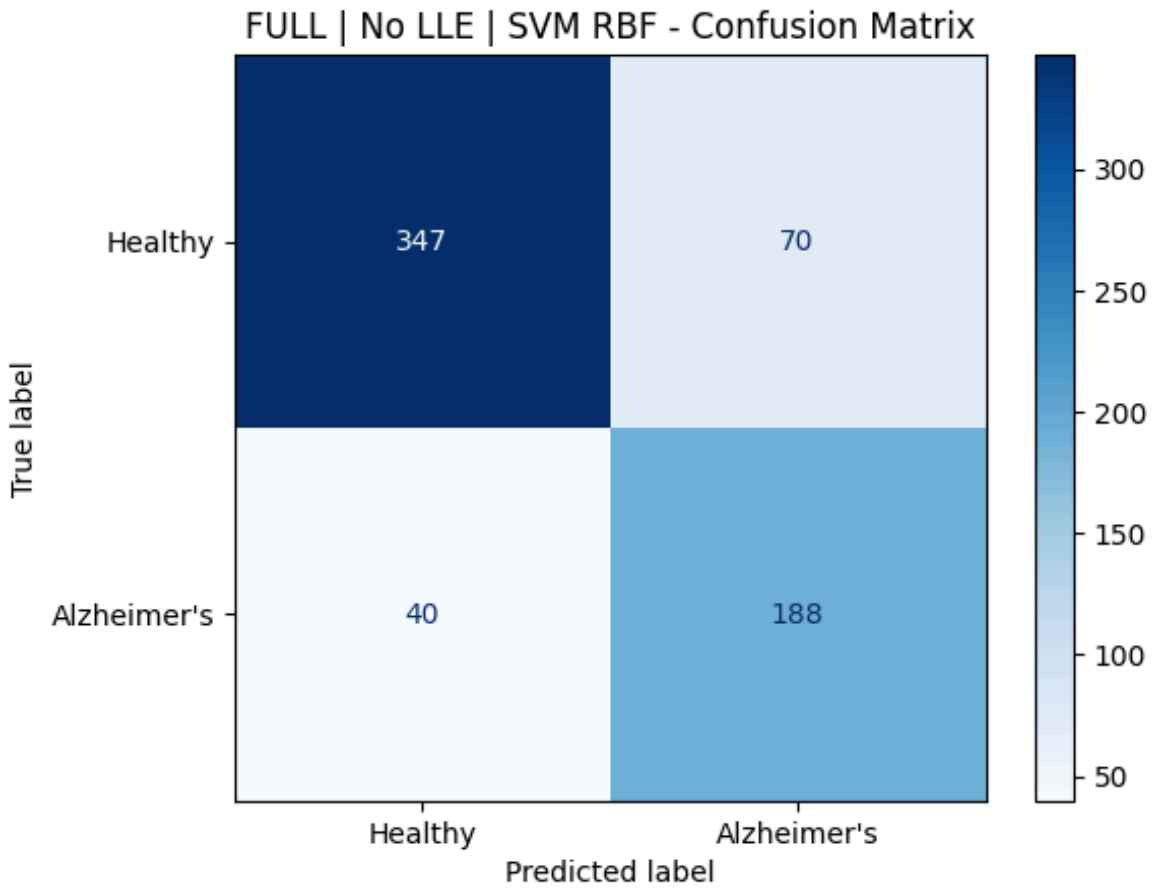

Figure 21. Full variables, no LLE + SVM RBF kernel: confusion matrix.

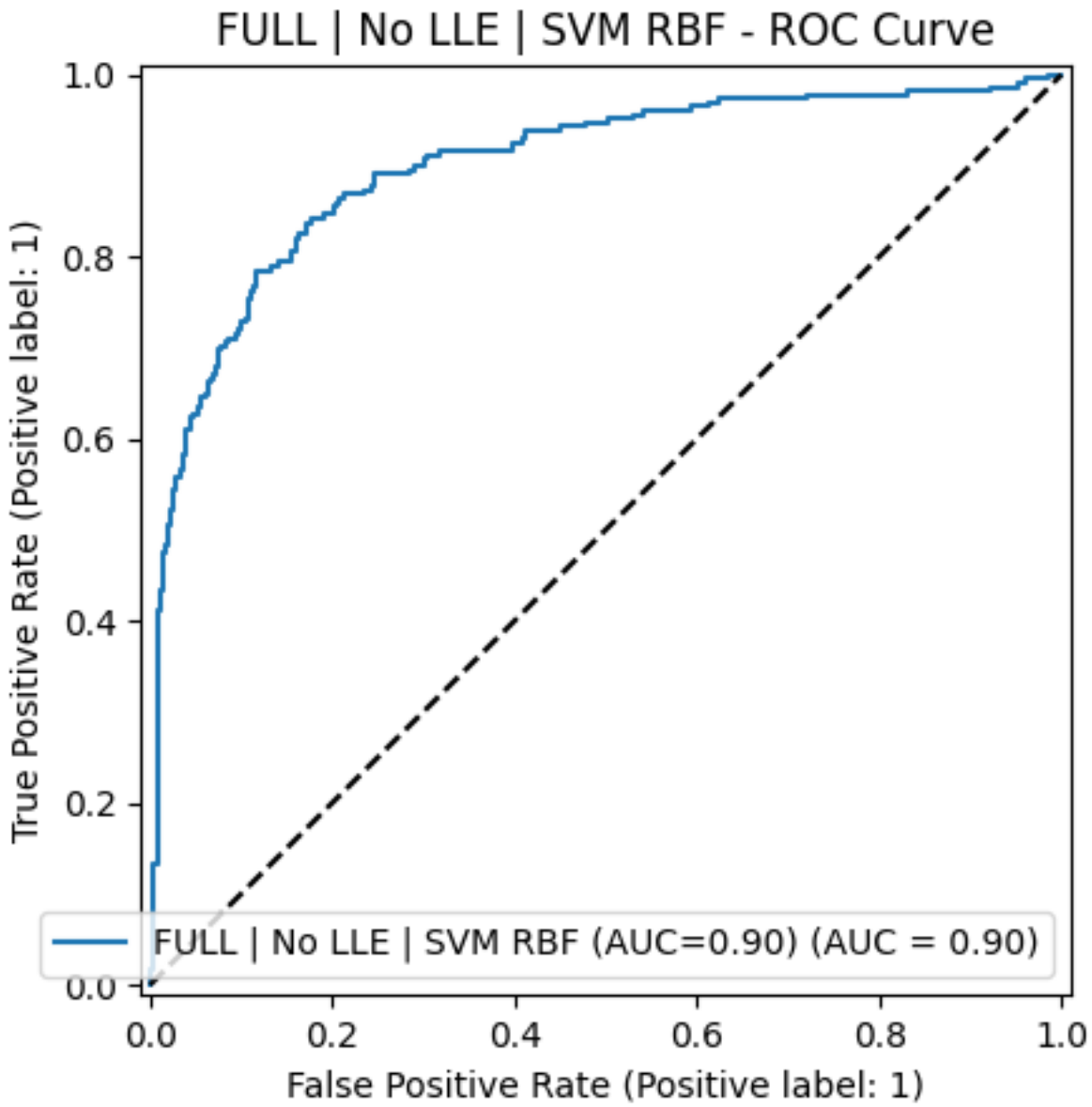

Figure 22. Full variables, no LLE + SVM RBF kernel: ROC AUC curve.

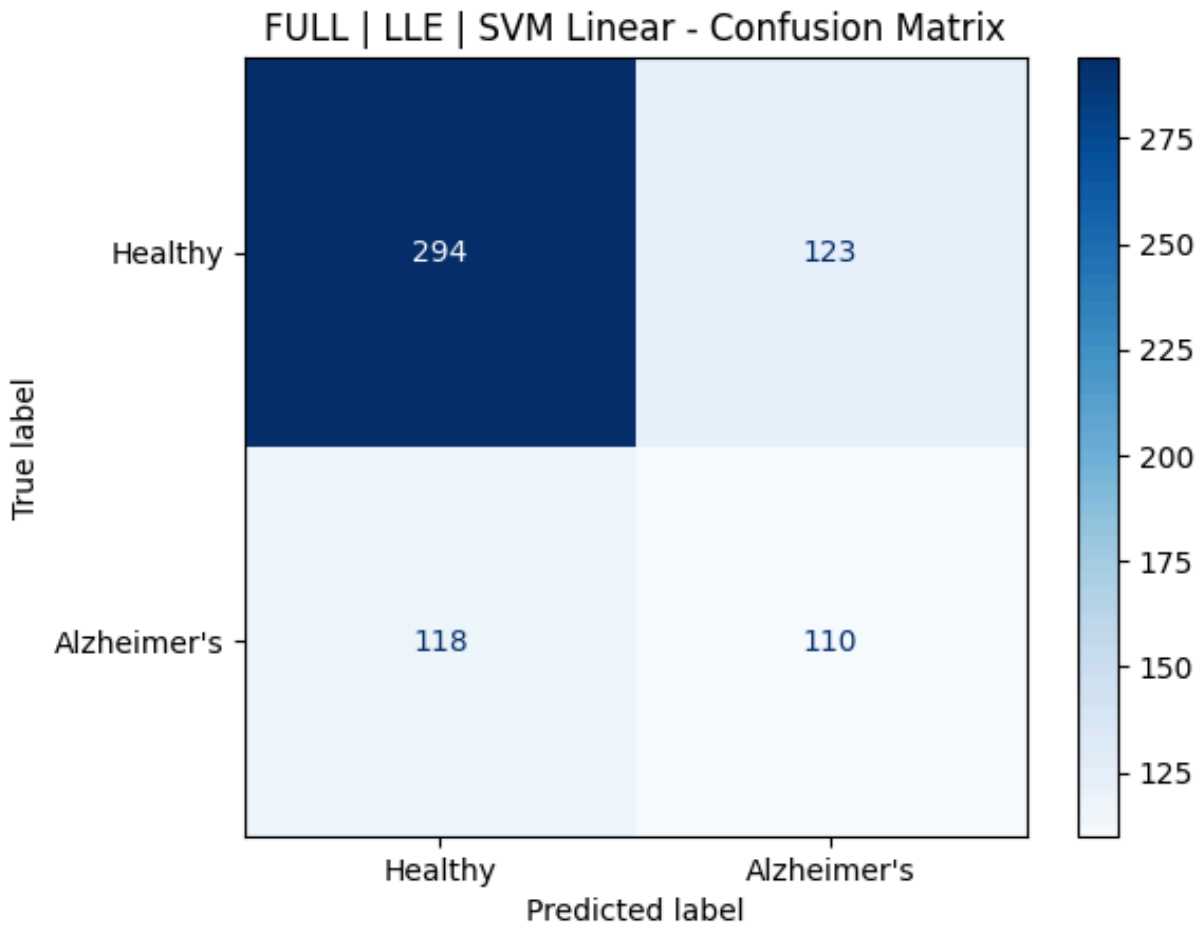

Figure 23. Full variables, LLE + SVM linear kernel: confusion matrix.

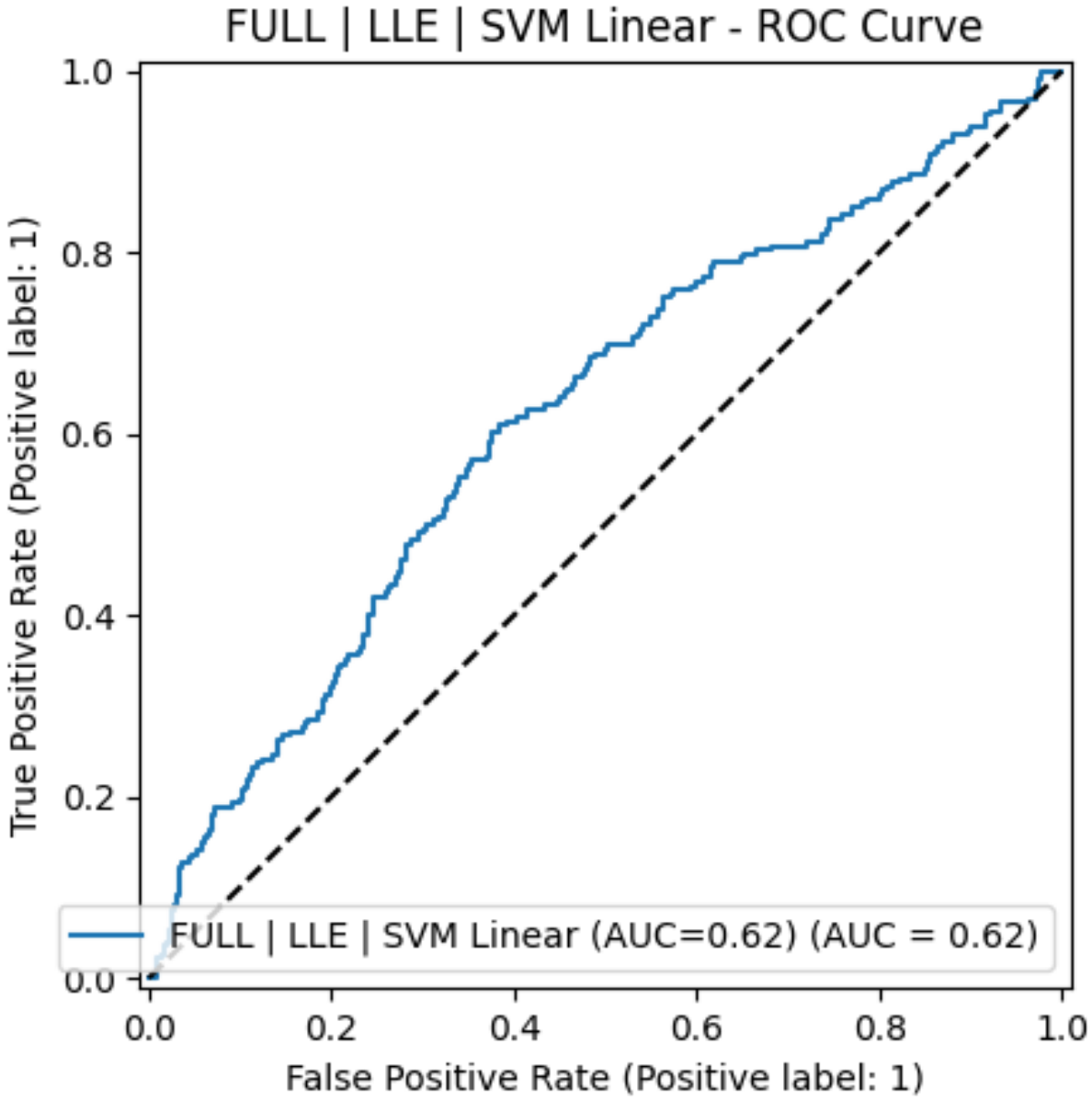

Figure 24. Full variables, no LLE + SVM RBF kernel: ROC AUC curve.

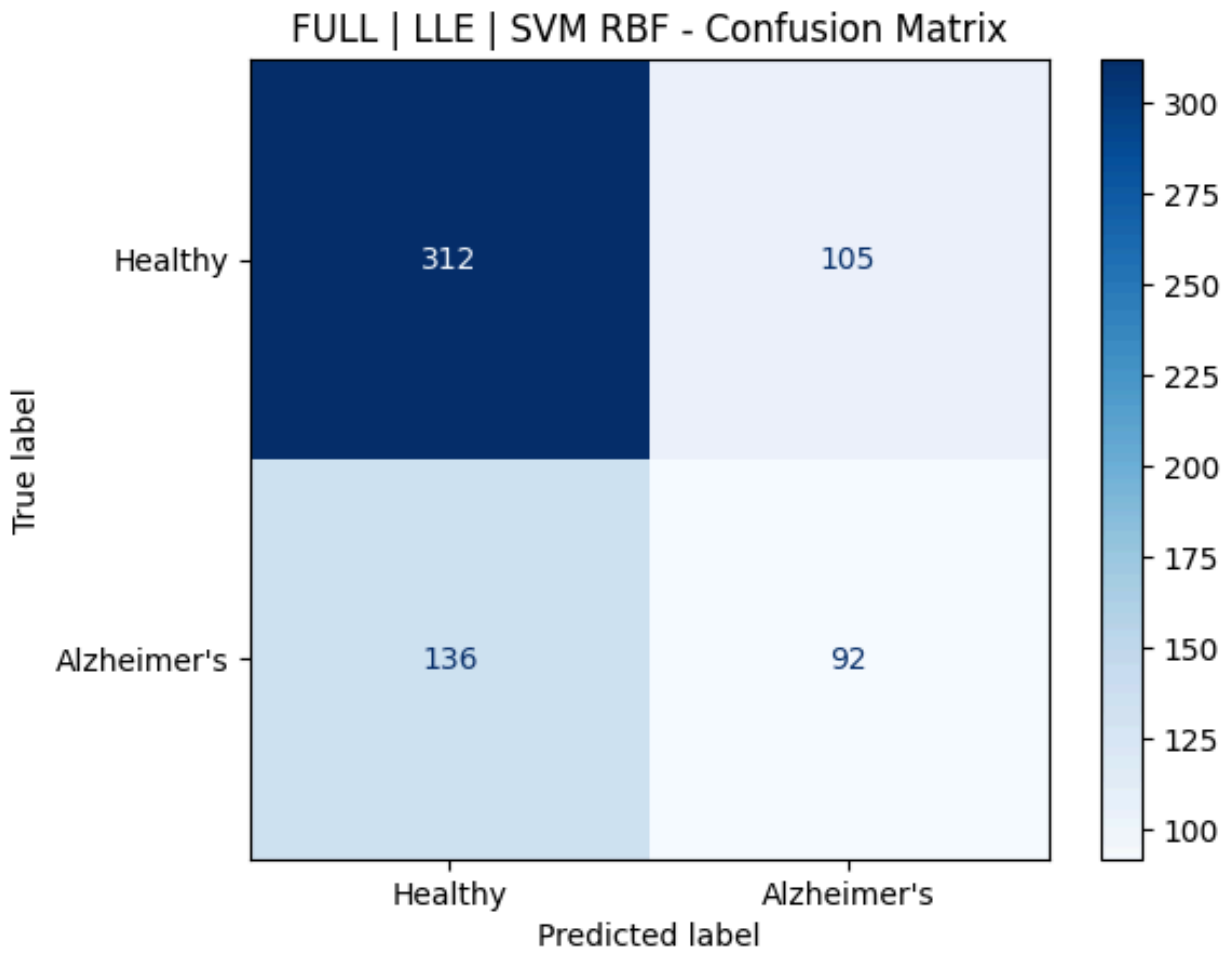

Figure 25. Full variables, LLE + SVM RBF kernel: confusion matrix.

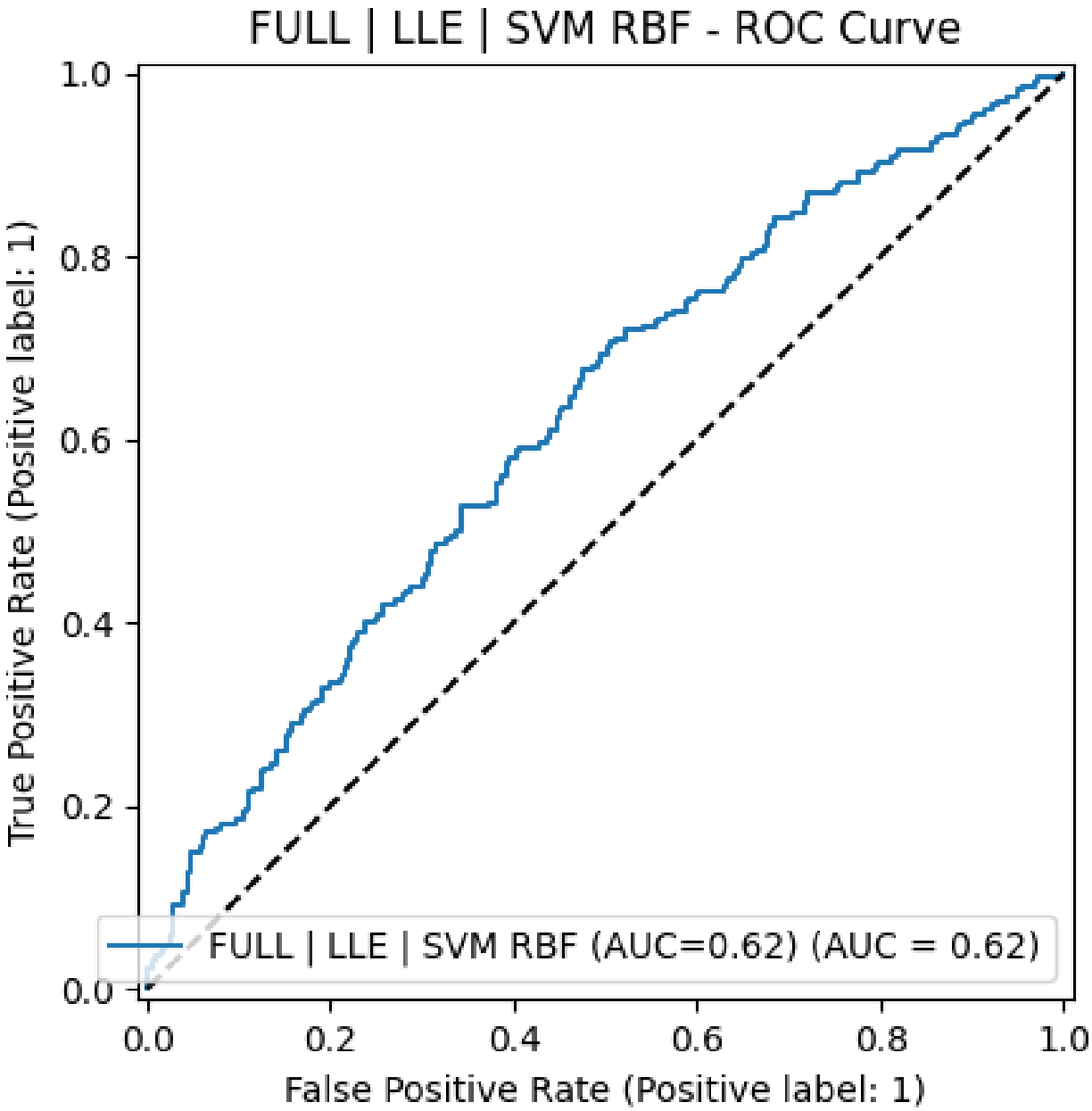

Figure 26. Full variables, LLE + SVM RBF kernel: ROC AUC curve.